TRIBHUVAN UNIVERSITY — SHANKER DEV CAMPUS (SDC)

BBA Eighth Semester — Preboard Examination 2026 Complete Combined Solutions
Includes All Questions, Theory, Step-by-Step Numericals, Diagrams & Exam Memory Hints

Subjects Covered in This Combined Solution PDF (All 4 Papers from SDC Preboard Examination 2026):
1. **BNK 202: Financial Derivatives** (Group A: Q1–Q10, Group B: Q11–Q17, Group C: Q18–Q21, Group D: Q22)
2. **BNK 221: Corporate Financing Decision** (Group A: Q1–Q10, Group B: Q11–Q17, Group C: Q18–Q21, Group D: Q22)
3. **BNK 209: Insurance and Risk Management** (Group A: Q1–Q10, Group B: Q11–Q17, Group C: Q18–Q21, Group D: Q22)
4. **MGT 240: Strategic Management** (Group A: Q1–Q10, Group B: Q11–Q17, Group C: Q18–Q21, Group D: Q22)

PAPER 1: SHANKER DEV CAMPUS (TU) — BNK 202: FINANCIAL DERIVATIVES

BBA / Eighth Semester | Preboard Examination 2026 | Full Marks: 100 | Time: 3 Hrs

Group A: Brief Answer Questions [10 x 2 = 20]

Q1. State the meaning of financial derivatives?

Answer: Financial derivatives are bilateral financial contracts (such as forwards, futures, options, and swaps) whose value is **derived from** the price or performance of an underlying asset, reference rate, index, or commodity. They are used primarily for hedging risk, speculation, and arbitrage.

Q2. What is the speculation value of call option?

Answer: The **speculative value (or time value)** of a call option is the excess of the option's market price (C_0) over its intrinsic value [max(S_0 - X, 0)]:
Speculative (Time) Value = Call Market Price (C_0) - Intrinsic Value [max(S_0 - X, 0)].
It reflects the value investors pay for the time remaining until expiration and the volatility/potential of the stock price rising further above the strike price.

Q3. Write four assumptions of binomial option pricing model?

Answer: Four key assumptions of the Binomial Option Pricing Model are:
• **1. Two Possible Price States per Period:** In each discrete time period, the underlying stock price can only move up by a multiplicative factor *u* or down by a factor *d*.
• **2. Frictionless Market:** There are no transaction costs, taxes, or margin restrictions, and short selling of assets is allowed.
• **3. Constant Risk-Free Rate (r):** Investors can borrow and lend at a known, constant risk-free interest rate *r*.
• **4. No-Arbitrage Condition:** Market prices adjust immediately to eliminate any riskless arbitrage opportunities (i.e., d < 1 + r < u).

Q4. What is the meaning of synthetic put?

Answer: A **synthetic put** is a portfolio constructed from the underlying stock, a call option, and risk-free lending that replicates the exact expiration payoff and profit profile of a put option via Put-Call Parity:
• **Synthetic Long Put = Long Call Option + Short Stock + Lending PV of Exercise Price [X * e^(-rT)]**.

Q5. Write about the bull spread strategy?

Answer: A **bull spread** is a vertical option spread strategy designed to profit from a moderate rise in the price of the underlying stock while limiting both maximum gain and maximum loss. It is created by **buying an option with a lower strike price (X_1) and simultaneously selling (writing) an option of the same type and expiration with a higher strike price (X_2 > X_1)** (using either two calls or two puts).

Q6. State the role of a clearing house in future trading.

Answer: In futures trading, the **clearinghouse** acts as the central counterparty to every trade (becoming the buyer to every seller and the seller to every buyer via novation). It guarantees contract performance, eliminates bilateral default risk, administers initial and maintenance margins, conducts daily marking-to-market settlement, and facilitates physical delivery or cash settlement.

Q7. A put option of ABC stock has an exercise price of Rs 150. The current price of stock is Rs 150. It is selling for a premium of Rs 12. The price of stock at expiration will be Rs 120. What is the gain/loss for put

seller?

Answer (Step-by-Step Numerical Solution):
Given:
• Exercise price (X) = Rs 150; Put premium received by seller (P) = Rs 12; Stock price at expiration (S_T) = Rs 120
Calculation of Gain/Loss for Put Seller (Writer):
• Payoff obligation of put seller at S_T = Rs 120: -max(X - S_T, 0) = -(150 - 120) = -Rs 30
• **Net Gain/Loss for Put Seller** = Put Premium Received - Payoff Obligation = Rs 12 - Rs 30 = **-Rs 18 (Loss of Rs 18 per share)**.

EXAM HINT TO REMEMBER: *Read carefully: Q7 asks for the PUT SELLER (Short Put), not the put buyer! Put Seller Profit = +Premium - max(X - S_T, 0) = +12 - (150 - 120) = -Rs 18 (a loss of Rs 18)!*

Q8. Suppose, the current price of the stock of XYZ Company is Rs 37, the exercise price of call is Rs 40, the risk-free interest rate is 5% per annum and the time to expiration is 0.5 year. Calculate the lower bound of European put.

Answer (Step-by-Step Numerical Solution):
Given: S_0 = Rs 37; X = Rs 40; r = 5% = 0.05; T = 0.5 year.
Formula for Lower Bound of a European Put (P_min):
• **Under Continuous Compounding:**
P_min = max[X * e^(-rT) - S_0 , 0] = 40 * e^(-0.05 * 0.5) - 37 = (40 * 0.97531) - 37 = 39.01 - 37 = **Rs 2.01**
• **Under Annual Compounding:**
P_min = max[X / (1 + r)^T - S_0 , 0] = 40 / (1.05)^0.5 - 37 = 39.04 - 37 = **Rs 2.04**

EXAM HINT TO REMEMBER: *Lower bound of European Put: P_min = max[X*e^(-rT) - S_0, 0] = 39.01 - 37 = Rs 2.01 (or Rs 2.04 with discrete compounding).*

Q9. Suppose the current price of the stock is Rs 30. A 3-month call on this stock is selling for Rs 7 and a 3 month put on same stock is selling Rs 3. The exercise price of both call and put is Rs 35. An investor buys a put option and a call option. What is the name of the strategy? What is breakeven price?

Answer (Step-by-Step Numerical Solution):
• **Name of the Strategy:** Buying 1 Call Option and 1 Put Option on the same stock with the same exercise price (X = Rs 35) and same expiration (3 months) is a **Long Straddle (Bottom Straddle)**.
• **Total Combined Premium Cost:** C + P = Rs 7 + Rs 3 = **Rs 10**.
• **Two Breakeven Stock Prices (BEP):**
1. **Lower Breakeven Price** = X - (C + P) = 35 - 10 = **Rs 25**
2. **Upper Breakeven Price** = X + (C + P) = 35 + 10 = **Rs 45**

EXAM HINT TO REMEMBER: *Long Straddle always has TWO breakeven prices: Strike - Total Premium (35 - 10 = Rs 25) and Strike + Total Premium (35 + 10 = Rs 45)!*

Q10. Consider a 10-month forward contract on a stock with a current price of Rs 50. The risk-free rate is 8 percent per annum with continuous compounding. The stock pays a dividend of Rs 0.75 per share after 3 months, 6 months and 9 months. Calculate the forward price.

Answer (Step-by-Step Numerical Solution):
Given: S_0 = Rs 50; r = 8% = 0.08 (continuous); T = 10/12 yr; Dividends D = Rs 0.75 at t_1 = 3/12, t_2 = 6/12, t_3 = 9/12 yr.
Step 1: Present Value of the Three Dividends (I):
I = 0.75 * e^(-0.08 * 3/12) + 0.75 * e^(-0.08 * 6/12) + 0.75 * e^(-0.08 * 9/12)
I = 0.75 * e^(-0.02) + 0.75 * e^(-0.04) + 0.75 * e^(-0.06)
I = 0.73515 + 0.72059 + 0.70632 = **Rs 2.1621 (or Rs 2.16)**
Step 2: 10-Month Forward Price (F_0):
F_0 = (S_0 - I) * e^(r * T) = (50 - 2.1621) * e^(0.08 * 10/12)
F_0 = 47.8379 * e^(0.066667) = 47.8379 * 1.068939 = **Rs 51.14**

EXAM HINT TO REMEMBER: *Dividend-Paying Stock Forward Price: First discount each dividend back to t=0 [I = Sum D_i*e^(-r*t_i) = Rs 2.1621], subtract I from S_0 (50 - 2.1621 = 47.8379), and compound forward for T = 10/12 year!*

Group B: Short Answer Questions (Attempt any SIX — ALL Solved) [6 x 5 = 30]

Q11. State and explain the major features of financial derivatives.

Answer: The major features of financial derivatives are:
• **1. Value Derived from an Underlying Asset:** A derivative has no standalone value; its price is determined by movements in an underlying stock, bond, currency, commodity, or interest rate.

• **2. Future Settlement & Contractual Nature:** Agreement is made today, while delivery or cash settlement occurs at a specified future date.
• **3. High Financial Leverage (Low Initial Capital):** Requires only a small margin or option premium upfront to control a large notional value.
• **4. Zero-Sum Game:** The realized gain of the long party equals the realized loss of the short party (excluding transaction costs).
• **5. Risk Transfer & Price Discovery:** Transfers price risk from hedgers to speculators and reveals consensus market expectations of future spot prices.

Q12. Explain why interest rate swaps are more widely used than currency and equity swaps?

Answer: Plain-vanilla **Interest Rate Swaps (IRS)** account for the vast majority (over 75%–80% of notional volume) of the global swap market compared to currency and equity swaps for the following reasons:
• **1. Universal Exposure to Interest Rate Risk:** Virtually every corporation, bank, government, and financial institution borrows or lends in its domestic currency and faces interest-rate risk (fixed vs. floating), whereas only multinational firms need currency swaps and mainly institutional portfolio managers use equity swaps.
• **2. Asset-Liability Management (ALM) by Commercial Banks:** Banks constantly use interest rate swaps to bridge the duration/repricing mismatch between their floating-rate deposits and fixed-rate loans.
• **3. No Exchange of Principal (Lower Credit/Default Risk):** In an interest rate swap, both legs are in the same currency, so notional principal is never exchanged—only small net interest differentials are settled. In contrast, currency swaps require exchanging huge principal amounts at maturity, creating much higher counterparty credit and FX settlement risk.
• **4. High Standardization, Liquidity, and Tight Bid-Ask Spreads:** Benchmark IRS contracts (tied to SOFR/EURIBOR) are highly liquid, centrally cleared, and carry minimal transaction costs.

Q13. Current stock price of Delta company is Rs 180 per share. A put option on this stock one month to expiration has exercise price of Rs 200 and selling for Rs 45. Another put option on same stock with same time to expiration has exercise price of Rs 190 and selling for Rs 30. Both options are American a. Examine whether they confirm the rules regarding American puts that differ only by exercise price b. Suggest arbitrage strategy if stock price at expiration (i) decreases to Rs 175, (ii) increases to Rs 225

Answer (Step-by-Step Numerical Solution):
Given:
• Current stock price (S_0) = Rs 180; Time to expiration = 1 month (Both puts are American)
• Lower-strike Put: X_1 = Rs 190, Put price $P(X_1)$ = Rs 30
• Higher-strike Put: X_2 = Rs 200, Put price $P(X_2)$ = Rs 45

a. Examination of Pricing Rules for Two American Puts Differing Only by Exercise Price ($X_2 > X_1$):
• **Rule 1 (Monotonicity Rule): $P(X_2) >= P(X_1)$**
A put with a higher exercise price must sell for at least as much as a put with a lower exercise price. Here, P(200) = Rs 45 >= P(190) = Rs 30. **(Rule 1 is Confirmed / Satisfied).**
• **Rule 2 (Maximum Spread / Slope Rule): $P(X_2) - P(X_1) <= X_2 - X_1$**
The difference in prices of two American puts cannot exceed the difference in their exercise prices.
- Actual Price Difference: P(200) - P(190) = Rs 45 - Rs 30 = **Rs 15**
- Exercise Price Difference: $X_2 - X_1$ = Rs 200 - Rs 190 = **Rs 10**
Since **Rs 15 > Rs 10**, **Rule 2 is VIOLATED!** The Rs 200-strike put is overpriced relative to the Rs 190-strike put.

b. Arbitrage Strategy and Profit at Expiration (for S_T = Rs 175 and S_T = Rs 225):
• **Arbitrage Strategy Today (Bear Put Spread Reversal / Bull Put Spread):**
- **Sell (Write) 1 overpriced Put (X_2 = Rs 200)** and receive **+Rs 45**
- **Buy 1 cheaper Put (X_1 = Rs 190)** and pay **-Rs 30**
- **Net Initial Cash Inflow Today (t = 0)** = +Rs 45 - Rs 30 = **+Rs 15 per share**
• **Net Arbitrage Profit at Expiration:**
- **(i) If Stock Price Decreases to S_T = Rs 175:**
• Long Put (X_1 = 190) Payoff = +(190 - 175) = +Rs 15
• Short Put (X_2 = 200) Payoff = -(200 - 175) = -Rs 25
• Net Expiration Payoff = +15 - 25 = -Rs 10
• **Net Arbitrage Profit** = Initial Inflow (+Rs 15) + Expiration Payoff (-Rs 10) = **+Rs 5 per share**.
- **(ii) If Stock Price Increases to S_T = Rs 225:**
• Both Put(190) and Put(200) are out-of-the-money and expire worthless (Payoff = Rs 0).
• **Net Arbitrage Profit** = Initial Inflow (+Rs 15) + Rs 0 = **+Rs 15 per share**.

EXAM HINT TO REMEMBER: *Put Spread Rule: The difference in put premiums [P(X_2) - P(X_1) = 45 - 30 = 15] can NEVER exceed the difference in strike prices [X_2 - X_1 = 200 - 190 = 10]! When violated, Sell the expensive put (+45) and Buy the cheap put (-30) for +Rs 15 upfront—your worst-case expiration outflow is -10, locking in at least +Rs 5 profit in every state!*

Q14. Consider a stock worth Rs 250 that can go up or down by 15 percent per period. The risk free rate is 10 percent. Use one binomial period. a. Determine the two possible stock prices for the next period. b. Determine the intrinsic values at expiration of a European call option with an exercise price of Rs 250. c. Find the value of the option today. d. What is the hedge portfolio? e. Show that the value of hedge portfolio will be the same for both stocks prices at

Answer (Step-by-Step Numerical Solution):
Given: S_0 = Rs 250; X = Rs 250; u = 1 + 0.15 = 1.15; d = 1 - 0.15 = 0.85; r = 10% = 0.10 (1 binomial period).

a. Two Possible Stock Prices Next Period:
• Up-state stock price (S_u) = 250 x 1.15 = **Rs 287.50**
• Down-state stock price (S_d) = 250 x 0.85 = **Rs 212.50**

b. Intrinsic Values (Payoffs) of the Call Option (X = Rs 250) at Expiration:
• In Up State: C_u = max(S_u - X, 0) = max(287.50 - 250, 0) = **Rs 37.50**
• In Down State: C_d = max(S_d - X, 0) = max(212.50 - 250, 0) = **Rs 0.00**

c. Value of the Call Option Today (C_0):
• Risk-neutral probability of up-move (p):
p = (1 + r - d) / (u - d) = (1.10 - 0.85) / (1.15 - 0.85) = 0.25 / 0.30 = **5/6 = 0.8333** (and 1 - p = 1/6 = 0.1667)
• Fair Call Value Today (C_0):
C_0 = [p * C_u + (1 - p) * C_d] / (1 + r) = [(5/6 x 37.50) + (1/6 x 0)] / 1.10 = 31.25 / 1.10 = **Rs 28.41**

d. The Hedge Portfolio (Delta Hedging Ratio, h):
• Hedge Ratio (h) = (C_u - C_d) / (S_u - S_d) = (37.50 - 0) / (287.50 - 212.50) = 37.50 / 75.00 = **0.50 shares per call**.
• Thus, the riskless **Hedge Portfolio** consists of **Buying (Long) h = 0.50 shares of stock and Writing (Short) 1 Call Option** (or equivalently, Buying 1 full share of stock and Writing 2 Call Options).

e. Proof That the Hedge Portfolio Value is Identical at Expiration in Both States:
• **If Stock Rises to S_u = Rs 287.50:**
Value of Hedge Portfolio (V_u) = h * S_u - C_u = (0.50 x 287.50) - 37.50 = 143.75 - 37.50 = **Rs 106.25**
• **If Stock Falls to S_d = Rs 212.50:**
Value of Hedge Portfolio (V_d) = h * S_d - C_d = (0.50 x 212.50) - 0 = **Rs 106.25**
(Note: Discounting Rs 106.25 at 10% gives today's hedge portfolio cost: 106.25 / 1.10 = Rs 96.59 = 0.50(250) - C_0 => C_0 = 125 - 96.59 = Rs 28.41!)

EXAM HINT TO REMEMBER: *In Q14(e), proving V_u = V_d is the best self-check in Binomial Pricing: 0.5(287.50) - 37.50 = Rs 106.25 and 0.5(212.50) - 0 = Rs 106.25! Because both equal Rs 106.25, the portfolio is 100% risk-free!*

Q15. You have provided the information about options on ABC stock: use Black-Scholes model [Stock variance = 0.16; Exercise price = Rs 110; Risk free rate = 10% p.a.; Current price of the stock = Rs 100; Time to expiration = 1 year; Dividend in 6 months = Rs 3] Required: What is the value of call option and put option?

Answer (Step-by-Step Numerical Solution):
Given:
• Current stock price (S_0) = Rs 100; Exercise price (X) = Rs 110
• **Stock Variance (sigma^2) = 0.16** => **Standard Deviation (sigma) = sqrt(0.16) = 0.40 (40%)**
• Risk-free rate (r) = 10% p.a. = 0.10; Time to expiration (T) = 1 year
• Dividend (D) = Rs 3 paid in 6 months (t = 0.5 year)

Step 1: Adjust Current Stock Price for Present Value of Dividend (S_0*):
• PV(Dividend) = D * e^(-r * t) = 3 * e^(-0.10 * 0.5) = 3 * e^(-0.05) = 3 * 0.951229 = **Rs 2.8537 (or Rs 2.85)**
• Adjusted Stock Price (S_0*) = S_0 - PV(Dividend) = 100 - 2.8537 = **Rs 97.1463 (or Rs 97.15)**

Step 2: Calculate d_1 and d_2 Using Adjusted Stock Price S_0* = Rs 97.1463:
d_1 = [ln(S_0* / X) + (r + 0.5 * sigma^2) * T] / (sigma * sqrt(T))
• ln(97.1463 / 110) = ln(0.883148) = **-0.124262**
• (r + 0.5 * sigma^2) * T = (0.10 + 0.5 * 0.16) * 1 = 0.10 + 0.08 = **0.18**
• sigma * sqrt(T) = 0.40 * sqrt(1) = **0.40**
• **d_1** = (-0.124262 + 0.18) / 0.40 = 0.055738 / 0.40 = **0.1393 (or 0.14)**
• **d_2** = d_1 - sigma * sqrt(T) = 0.1393 - 0.40 = **-0.2607 (or -0.26)**

Step 3: Cumulative Normal Probabilities from Z-Table:
• N(d_1) = N(0.1393) = **0.5554** *(At Z = 0.14: N(0.14) = 0.5557)*
• N(d_2) = N(-0.2607) = **0.3972** *(At Z = -0.26: N(-0.26) = 0.3974)*

Step 4: Value of Call Option (C_0) and Put Option (P_0):
• Call Option Value (C_0):
C_0 = S_0* * N(d_1) - X * e^(-rT) * N(d_2)
C_0 = (97.1463 x 0.5554) - (110 x e^(-0.10) x 0.3972)
C_0 = 53.955 - (99.532 x 0.3972) = 53.955 - 39.534 = **Rs 14.42**
• Put Option Value (P_0) via Dividend-Adjusted Put-Call Parity:
P_0 = C_0 + X * e^(-rT) - S_0*
P_0 = 14.42 + 99.53 - 97.15 = **Rs 16.81 (or Rs 16.80)**

EXAM HINT TO REMEMBER: *Watch out for TWO traps in Q15: (1) The table gives Stock VARIANCE = 0.16, so sigma^2 is ALREADY 0.16 and sigma = sqrt(0.16) = 0.40! Do NOT square 0.16 again! (2) Subtract PV of the Rs 3 dividend [3*e^(-0.05) = Rs 2.85] from S_0 = 100 to get S_0* = Rs 97.15 before plugging into BSM and Put-Call Parity!*

Q16. An investor enters into a short future contract at the opening of the market to sell 5,000 bushels of wheat for Rs 2.5 per bushels. The initial margin is Rs 3,000 and the maintenance margin is Rs 2,000. a. What price change would lead to a margin call? Calculate the margin call price. b. Under what circumstances could Rs 1,500 be withdrawn from the margin account?

Answer (Step-by-Step Numerical Solution):
Given:
• Position: **Short Futures Contract** for Q = 5,000 bushels of wheat
• Initial futures price (F_0) = Rs 2.50 per bushel
• Initial Margin (IM) = Rs 3,000; Maintenance Margin (MM) = Rs 2,000

a. Price Change Leading to a Margin Call and Margin Call Price:
• Maximum allowable loss before margin call = IM - MM = Rs 3,000 - Rs 2,000 = **Rs 1,000**
• Because the investor has a **SHORT position**, losses occur when the wheat futures price **RISES**:
Price Increase per bushel = Allowable Loss / Contract Size = Rs 1,000 / 5,000 bushels = **+Rs 0.20 per bushel**
• **Margin Call Price** = F_0 + Rs 0.20 = Rs 2.50 + Rs 0.20 = **Rs 2.70 per bushel** (any price above Rs 2.70 triggers a margin call).

b. Circumstances Under Which Rs 1,500 Could Be Withdrawn from the Margin Account:
• To withdraw Rs 1,500 above the Initial Margin, the short position must earn a gain of Rs 1,500 (which happens when the wheat futures price **FALLS**):
Required Price Decrease = Rs 1,500 / 5,000 bushels = **-Rs 0.30 per bushel**
• Thus, Rs 1,500 can be withdrawn if the futures price **decreases by Rs 0.30 per bushel to Rs 2.50 - Rs 0.30 = Rs 2.20 per bushel (or lower)**.

EXAM HINT TO REMEMBER: *For a SHORT futures position: Margin Call happens when price RISES [F_0 + (IM - MM)/Q = 2.50 + 0.20 = Rs 2.70], and Withdrawal happens when price FALLS [F_0 - Withdrawal/Q = 2.50 - 0.30 = Rs 2.20]!*

Q17. Consider an asset priced at $50. The risk-free interest rate is 8 percent, and a futures contract on the asset expires in 45 days. Answer the following, with questions, each of which is independent of the others, unless indicated otherwise. a. Find the appropriate futures price if the underlying asset has no storage costs, cash flows, or convenience yield. b. Find the appropriate futures price if the future value of storage costs on underlying assets at the futures expiration equals $2.25 c. Find the appropriate futures price if the underlying asset has positive cash flows. The future value of these cash flows is $0.75 at the time of future expiration. d. Find the appropriate futures price if the future value of the net overall cost of carry on the underlying assets equals $3.55 e. Using Part D above, illustrate how an arbitrage transaction could be executed if the futures contract is trading at $60.

Answer (Step-by-Step Numerical Solution):
Given: Spot price (S_0) = $50; Risk-free rate (r) = 8% = 0.08; Time to expiration (T) = 45 days.

a. Futures Price with No Storage Costs, Cash Flows, or Convenience Yield:
• Using Discrete Annual Compounding (365-day year — Standard Don Chance Formula):
F_0 = S_0 * (1 + r)^(45 / 365) = 50 * (1.08)^(0.123288) = 50 * 1.009532 = **$50.48**
*(Note: Under continuous/simple interest: 50 * e^(0.08 * 45/365) = **$50.50**.)*

b. Futures Price When Future Value of Storage Costs at Expiration (FV_SC) = $2.25:
Storage costs are an additional cost of carrying the asset and increase the futures price:

F_0 = S_0 * (1 + r)^T + FV(Storage Costs) = $50.48 + $2.25 = **$52.73** (or $50.50 + $2.25 = **$52.75**).

c. Futures Price When Future Value of Positive Cash Flows at Expiration (FV_CF) = $0.75:
Positive cash flows (dividends/coupons) reduce the net cost of carry and lower the futures price:
F_0 = S_0 * (1 + r)^T - FV(Cash Flows) = $50.48 - $0.75 = **$49.73** (or $50.50 - $0.75 = **$49.75**).

d. Futures Price When Future Value of Net Overall Cost of Carry = $3.55:
• Since the Future Value of the **net overall cost of carry** (interest + storage - cash flows/convenience yield) is $3.55:
F_0 = S_0 + Net Overall Cost of Carry = $50.00 + $3.55 = $53.55
*(Note: If $3.55 represents net non-interest carry [storage minus cash flows], then F_0 = $50.48 + $3.55 = **$54.03**.)*

e. Arbitrage Transaction Using Part (d) When Market Futures Price (F_mkt) = $60.00:
• Since Market Futures Price ($60.00) > Fair Futures Price ($53.55), the futures contract is **overpriced** -> Execute **Cash-and-Carry Arbitrage**:
1. **Today (t = 0):** Borrow $50.00 at the risk-free rate, buy the underlying asset in the spot market for $50.00, and **sell (short) the 45-day futures contract** at $60.00.
2. **At Expiration (Day 45):** Deliver the asset under the short futures contract to receive **$60.00**, and pay the original loan plus the net overall cost of carry totaling **$53.55** (or $54.03).
3. **Riskless Arbitrage Profit at Expiration:** $60.00 - $53.55 = **$6.45 per unit** (or $60.00 - $54.03 = **$5.97 per unit**).

EXAM HINT TO REMEMBER: *Cost-of-Carry Golden Rule: Storage Costs ADD to Futures Price (+$2.25), while Cash Flows/Dividends SUBTRACT from Futures Price (-$0.75). Overall Futures Price = Spot Price + Net Cost of Carry ($50 + $3.55 = $53.55)!*

Group C: Long Answer Questions (Attempt any THREE — ALL Solved) [3 x 10 = 30]

Q18. What is the risk management? Discuss the rationales for risk management.

Answer:
1. Meaning of Risk Management:
In corporate finance and derivatives, risk management is the systematic process of identifying and measuring the financial risks faced by a firm (such as foreign exchange risk, interest rate risk, commodity price risk, and equity price risk) and using financial instruments (forwards, futures, options, swaps) or operational techniques to adjust the firm's risk profile to a desired target level that maximizes firm value.

2. Rationales for Corporate Risk Management (Why Firms Hedge in Imperfect Markets):
Under Modigliani and Miller's perfect-market assumptions, shareholders could diversify risk on their own. However, in the real world of market imperfections, corporate risk management directly increases firm value for the following major rationales:
• **1. Reduction of Expected Costs of Financial Distress and Bankruptcy:**
Unhedged volatility in cash flows can push a firm into financial distress, triggering direct legal/administrative bankruptcy costs and indirect costs (loss of customers, suppliers refusing trade credit, key employees leaving). Hedging stabilizes cash flows and lowers the probability of distress.
• **2. Increasing Debt Capacity and Interest Tax Shields:**
Because hedging reduces cash-flow volatility and default risk, lenders are willing to lend more at lower interest rates. Higher safe leverage allows the firm to capture larger tax shields on debt interest.
• **3. Avoiding the Underinvestment Problem (Preserving Internal Funds for Positive-NPV Projects):**
As shown by Froot, Scharfstein, and Stein, external financing is costly when cash flows drop. Risk management ensures the firm always has sufficient internal cash flows to fund valuable R&D and capital investment opportunities without cutting capex.
• **4. Convexity of the Corporate Tax Schedule (Reducing Expected Taxes):**
When a firm faces a progressive tax rate schedule or limits on tax-loss carryforwards, volatile pre-tax income leads to higher average tax payments than stable pre-tax income. Hedging smooths taxable income and lowers expected taxes.
• **5. Protecting Undiversified Stakeholders (Managers, Employees, Suppliers):**
Unlike institutional shareholders, managers and employees have their human capital and wealth concentrated in the firm; reducing firm risk lowers the risk premium they demand in compensation.
• **6. Better Performance Evaluation and Budgeting Accuracy:**
Hedging external macro noise (like currency or oil shocks) allows the board to evaluate managers based on true operational performance and makes capital budgeting more accurate.

Q19. An investor buys a call and a put on the stock of ABC Company. Both put and call matures 3 months hence. The exercise price of call is Rs 45 and selling at Rs 5. The exercise price of put is Rs 35 and selling at Rs 3. a. What is the name of the position created by investors? b. Find out the total costs of position? c.

Calculate the gain or loss of the strategy if ending prices becomes Rs 15, Rs 20, Rs 25, Rs 30, Rs 35, Rs 40, Rs 45, Rs 50, Rs 55, Rs 60 and Rs 65. d. Calculate the maximum gain and loss for the position e. What are the breakeven prices?

Answer (Step-by-Step Numerical Solution):

Given:

• Call Option: Exercise price (X_C) = Rs 45; Call premium (C) = Rs 5

• Put Option: Exercise price (X_P) = Rs 35; Put premium (P) = Rs 3; Maturity = 3 months

a. Name of the Position Created by the Investor:

Buying 1 Call Option and 1 Put Option with the same expiration date but **different exercise prices (X_P = Rs 35 < X_C = Rs 45)** is called a **Long Strangle** (or Bottom Vertical Combination).

b. Total Cost of the Position:

Total Cost = Call Premium (C) + Put Premium (P) = Rs 5 + Rs 3 = **Rs 8 per share**.

c. Gain or Loss Schedule for Ending Stock Prices from Rs 15 to Rs 65:

Ending Stock Price (S_T)	Call Payoff max(S_T - 45, 0)	Put Payoff max(35 - S_T, 0)	Combined Payoff (Rs)	Total Initial Cost C + P (Rs)	Net Gain / Loss (Payoff - 8)
Rs 15	Rs 0	Rs 20	Rs 20	Rs 8	**+Rs 12 (Gain)**
Rs 20	Rs 0	Rs 15	Rs 15	Rs 8	**+Rs 7 (Gain)**
Rs 25	Rs 0	Rs 10	Rs 10	Rs 8	**+Rs 2 (Gain)**
Rs 30	Rs 0	Rs 5	Rs 5	Rs 8	-Rs 3 (Loss)
Rs 35 (Put Strike)	Rs 0	Rs 0	Rs 0	Rs 8	**-Rs 8 (Max Loss)**
Rs 40	Rs 0	Rs 0	Rs 0	Rs 8	**-Rs 8 (Max Loss)**
Rs 45 (Call Strike)	Rs 0	Rs 0	Rs 0	Rs 8	**-Rs 8 (Max Loss)**
Rs 50	Rs 5	Rs 0	Rs 5	Rs 8	-Rs 3 (Loss)
Rs 55	Rs 10	Rs 0	Rs 10	Rs 8	**+Rs 2 (Gain)**
Rs 60	Rs 15	Rs 0	Rs 15	Rs 8	**+Rs 7 (Gain)**
Rs 65	Rs 20	Rs 0	Rs 20	Rs 8	**+Rs 12 (Gain)**

d. Maximum Gain and Maximum Loss for the Position:

• **Maximum Possible Gain: Unlimited on the upside** (as S_T -> Infinity, profit -> Infinity); on the downside (if S_T falls to Rs 0), maximum downside gain = Rs 35 - Rs 8 = **Rs 27 per share**.

• **Maximum Possible Loss: Rs 8 per share** (occurs whenever S_T lies between Rs 35 and Rs 45 inclusive).

e. Breakeven Stock Prices (BEP):

• **Lower Breakeven Price (BEP_1)** = Put Strike (X_P) - Total Premium = Rs 35 - Rs 8 = **Rs 27**

• **Upper Breakeven Price (BEP_2)** = Call Strike (X_C) + Total Premium = Rs 45 + Rs 8 = **Rs 53**

EXAM HINT TO REMEMBER: *Difference between Straddle and Strangle: Same strike (X_C = X_P) = Straddle; Different strikes (X_P = 35 < X_C = 45) = Strangle! For a Long Strangle, Lower BEP = X_P - Cost = 35 - 8 = Rs 27, and Upper BEP = X_C + Cost = 45 + 8 = Rs 53!*

Q20. An American put option on stock of City Bank has Rs 600 exercise price. The stock is currently selling for Rs 580. The option expires after one-year and there are two binomial periods in a year. At the end of each binomial period, either the stock price increases by 20 percent or decreases by 20 percent. The stock pays Rs 50 dividend at the end of first period. The risk-free rate of interest is 10 percent per annum. Find the value of put option today a. Construct binomial price tree and calculate possible stock prices at each node. b. Determine the possible prices of the put at expiration. c. Calculate the probability of a rise d. Find the possible prices of the put at the end of first period. e. What is the current price of the put option? f. Suppose the actual call [put] is selling for Rs 80. Suggest a strategy.

Answer (Step-by-Step Numerical Solution):

Given:

• Current stock price (S_0) = Rs 580; Exercise price (X) = Rs 600 (2-Period American Put Option)

• Up factor (u) = 1.20; Down factor (d) = 0.80

• Dividend at end of Period 1 (D_1) = Rs 50

• Annual risk-free rate = 10% p.a. => **Periodic risk-free rate per 6-month period (r_p) = 10% / 2 = 5% = 0.05**

(Note: If 10% is used per period, those figures are also given in parentheses below.)

a. Two-Period Binomial Stock Price Tree (With Rs 50 Ex-Dividend Drop at Period 1):
• **Period 0 (Today):** S_0 = **Rs 580.00**
• **Period 1 (End of 6 Months — Cum-Dividend and Ex-Dividend):**
- Up Node (U): Cum-dividend S_u(cum) = 580 x 1.20 = Rs 696.00 => **Ex-dividend S_u = 696 - 50 = Rs 646.00**
- Down Node (D): Cum-dividend S_d(cum) = 580 x 0.80 = Rs 464.00 => **Ex-dividend S_d = 464 - 50 = Rs 414.00**
• **Period 2 (At Expiration — 4 Non-Recombining Nodes from Ex-Dividend Period 1 Prices):**
- Node UU: S_uu = 646 x 1.20 = **Rs 775.20**
- Node UD: S_ud = 646 x 0.80 = **Rs 516.80**
- Node DU: S_du = 414 x 1.20 = **Rs 496.80**
- Node DD: S_dd = 414 x 0.80 = **Rs 331.20**

b. Possible Prices (Payoffs) of the Put Option (X = Rs 600) at Expiration (Period 2):
• P_uu = max(600 - 775.20, 0) = **Rs 0.00**
• P_ud = max(600 - 516.80, 0) = **Rs 83.20**
• P_du = max(600 - 496.80, 0) = **Rs 103.20**
• P_dd = max(600 - 331.20, 0) = **Rs 268.80**

c. Risk-Neutral Probability of a Rise (p):
• Using r_p = 5% per 6-month period:
p = (1 + r_p - d) / (u - d) = (1.05 - 0.80) / (1.20 - 0.80) = 0.25 / 0.40 = **0.625** (and 1 - p = **0.375**)
*(Note: If r_p = 10% per period is used: p = (1.10 - 0.80) / 0.40 = **0.75** and 1 - p = **0.25**.)*

d. Possible Prices of the American Put at the End of First Period (Checking Early Exercise):
• **At Node U (S_u,ex = Rs 646):**
- Hold value: P_u(hold) = [0.625(0) + 0.375(83.20)] / 1.05 = 31.20 / 1.05 = **Rs 29.71**
- Early exercise value: max(600 - 646, 0) = Rs 0
- Therefore, **P_u = max(29.71, 0) = Rs 29.71** *(At r_p = 10%: P_u = [0.25 x 83.20]/1.10 = **Rs 18.91**)*
• **At Node D (S_d,ex = Rs 414):**
- Hold value: P_d(hold) = [0.625(103.20) + 0.375(268.80)] / 1.05 = (64.50 + 100.80) / 1.05 = 165.30 / 1.05 = **Rs 157.43**
- Early exercise value (ex-dividend): P_d(exercise) = 600 - 414 = **Rs 186.00**
- Since Rs 186.00 > Rs 157.43, the American put **IS exercised early at Node D**!
- Therefore, **P_d = max(157.43, 186.00) = Rs 186.00** *(At r_p = 10%: P_d(hold) = 131.45, so P_d = **Rs 186.00**)*

e. Current Price of the American Put Option Today (P_0):
• P_0(hold) = [p * P_u + (1 - p) * P_d] / (1 + r_p)
P_0(hold) = [0.625(29.714) + 0.375(186.00)] / 1.05 = (18.571 + 69.750) / 1.05 = 88.321 / 1.05 = **Rs 84.12**
• Immediate exercise today: max(600 - 580, 0) = Rs 20.00
• Since Rs 84.12 > Rs 20.00, **Current Value of the American Put (P_0) = Rs 84.12**
*(Note: For European Put without early exercise, P_0 = **Rs 73.91**; if r_p = 10% per period, American Put P_0 = **Rs 55.17**.)*

f. Investment Strategy When Actual Option is Selling for Rs 80:
Since the actual market price (**Rs 80.00**) is less than the fair theoretical American put price (**Rs 84.12**), the option is **underpriced by Rs 4.12** -> **BUY the Put Option** (and buy the delta hedge ratio of the stock to lock in riskless arbitrage). *(If using the r_p = 10% value of Rs 55.17, then Rs 80 is overpriced -> Sell/Write the option.)*

***EXAM HINT TO REMEMBER:** When a discrete rupee dividend (Rs 50) is paid at Period 1, S_u and S_d drop by Rs 50 (to 646 and 414), so S_ud (516.80) != S_du (496.80)—meaning Period 2 has FOUR separate nodes! Always check early exercise at Node D: 600 - 414 = Rs 186 > Hold value Rs 157.43!*

Q21. A currency swap with semiannual payments for two years between hypothetical U.S. and Global Swaps, Inc. was entered into on 1 Jan 2021. Both counterparties agreed to pay fixed rate. The notional principals were EUR 10 million and $9.804 million which were equivalent with exchange rate of swap initiation date. Assuming 30-day month and 360-day year. [Init Term Structure: 180d: USD 5.5%, EUR 3.8%; 360d: USD 5.5%, EUR 4.2%; 540d: USD 6.2%, EUR 4.4%; 720d: USD 6.4%, EUR 4.5%] a. Calculate the fixed rate in dollar and euro. b. Assume it is 90 days into the life of swap. The exchange rate now is $0.9790/EUR and the term structures are as given below: [New Term Structure at 90d: 90d: USD 5.7%, EUR 3.9%; 270d: USD 6.1%, EUR 4.3%; 450d: USD 6.4%, EUR 4.5%; 630d: USD 6.6%, EUR 4.6%] Find the value of swap. c. Swap contract is the important financial instrument to exchange pattern of payment. In this regard, describe the motives of swap.

Answer (Step-by-Step Numerical & Theoretical Solution):

a. Calculation of Fixed Rates in US Dollar and Euro at Initiation (t = 0):
• **Discount Factors at Initiation B_i = 1 / [1 + R_i * (Days_i / 360)]:**
- **USD Discount Factors:** B_1(180d) = 0.973236; B_2(360d) = 0.947867; B_3(540d) = 0.914913; B_4(720d) = 0.886525 =>

Sum B_USD = 3.722541
- **EUR Discount Factors:** B_1(180d) = 0.981354; B_2(360d) = 0.959693; B_3(540d) = 0.938086; B_4(720d) = 0.917431 => **Sum B_EUR = 3.796564**

• Semiannual and Annualized Fixed Swap Rates:
- **USD Fixed Rate:** Semiannual c_USD = (1 - 0.886525) / 3.722541 = **3.0483% per half-year** => **Annualized R_USD = 3.0483% x 2 = 6.0967% (6.10% p.a.)**
- **EUR Fixed Rate:** Semiannual c_EUR = (1 - 0.917431) / 3.796564 = **2.1748% per half-year** => **Annualized R_EUR = 2.1748% x 2 = 4.3497% (4.35% p.a.)**

b. Valuation of the Currency Swap 90 Days into the Life of the Swap (Spot = $0.9790/EUR):

• New Discount Factors at Day 90 (for 90, 270, 450, 630 days):
- **New USD Discount Factors (B'_USD):** B'_1 = 0.985950; B'_2 = 0.956252; B'_3 = 0.925926; B'_4 = 0.896459 => **Sum B'_USD = 3.764587**
- **New EUR Discount Factors (B'_EUR):** B'_1 = 0.990344; B'_2 = 0.968758; B'_3 = 0.946746; B'_4 = 0.925497 => **Sum B'_EUR = 3.831345**

• Present Value of USD Leg and EUR Leg at Day 90:
- Value of USD Leg per $1 = (0.030483 x 3.764587) + 0.896459 = $1.011215 => **Total V_USD = $9,804,000 x 1.011215 = $9,913,960**
- Value of EUR Leg per EUR 1 = (0.021748 x 3.831345) + 0.925497 = EUR 1.008821 => **Total V_EUR = EUR 10,088,227**
- Converted into USD at S_90 = $0.9790/EUR: **V_EUR(in USD) = EUR 10,088,227 x 0.9790 = $9,876,374**

• Net Value of Swap at Day 90:
- To U.S. Firm (Receives USD, Pays EUR): $9,913,960 - $9,876,374 = **+$37,586**
- To Global Swaps, Inc. (Pays USD, Receives EUR): **-$37,586**

c. Motives for Entering into Swap Contracts:

• **1. Exploiting Comparative Advantage to Lower Borrowing Costs:** One firm may have a comparative advantage in fixed-rate (or home-currency) credit markets while another has a comparative advantage in floating-rate (or foreign-currency) markets; a swap allows both firms to borrow where they have an advantage and lower their borrowing rates.

• **2. Hedging Interest Rate and Foreign Exchange Risk:** Matching the currency and interest nature (fixed vs. floating) of liabilities with the currency and nature of asset cash inflows (Asset-Liability Management).

• **3. Overcoming Capital Market Access Barriers & Exchange Controls:** Obtaining long-term foreign-currency financing in markets where a foreign firm is less known.

• **4. Lower Transaction Costs and Privately Tailored Maturities:** Restructuring an entire balance sheet's debt profile with a single OTC swap contract at much lower cost than redeeming and reissuing bonds.

Group D: Comprehensive Answer Question [20 Marks]

Q22. Case Study: Metro Bank Limited (MBL) Stock & Options Today's closing price of MBL stock is S_0 = Rs 150. 6-month Call options with Exercise Price X = Rs 150 sell for C = Rs 10, and 6-month Put options with Exercise Price X = Rs 150 sell for P = Rs 5. a. If you believe that the political condition will settle soon and you want to speculate in stock, what will you do? How much will you gain/loss if stock price after six months increased to Rs 200 per share? b. If you believe that the political condition will not settle soon and you want to invest in stock, what will you do? How much will you gain/loss if stock price after six months decreased to Rs 100 per share? c. If you believe that the political condition will settle soon but you want to speculate in options, what will you do? How much will you gain/loss if stock price after six months increased to Rs 200 per share? At what stock price after 6 months will you make breakeven? d. If you believe that the political condition will not settle soon but you want to speculate options, what will you do? How much will you gain/loss if stock price after six months decreased to Rs 100 per share? At what stock price after 6 months will you make breakeven? e. If you believe that the political condition will settle soon and assume that you cannot use the call option, what will be your investment strategy? How much will you gain/loss if stock price after six months increased to Rs 200 per share? At what stock price after 6 months will you make breakeven? f. If you invested in stock as described in part 'a' but the political condition become worse as oppose your expectation. Now you would like to hedge the stock position with put option. Which strategy will you follow? Show the profit/loss from your strategy at expiration for Notional prices of Rs 130 to Rs 170 with Rs 10 interval. What stock price at expiration make you breakeven? Plot the graph of your strategy. g. If you invested in stock as described in part 'a' but the political condition become worse as oppose your expectation. Now you would like to hedge the stock position with call option. Which strategy will you follow? Show the profit/loss from your strategy at expiration for Notional prices of Rs 130 to Rs 170 with Rs 10 interval. What stock price at expiration will make you breakeven? Plot the graph of your strategy.

Answer (Comprehensive Case Study Solution):
Given Summary: MBL Current Stock Price (S_0) = Rs 150; Strike Price (X) = Rs 150; 6-Month Call Premium (C) = Rs 10; 6-Month Put Premium (P) = Rs 5.

a. Bullish Speculation in Stock (Political Condition Settles Soon):
• **Strategy: Buy MBL Stock (Long Stock Position)** at S_0 = Rs 150 per share.
• **Gain at S_T = Rs 200:** Profit = S_T - S_0 = Rs 200 - Rs 150 = **+Rs 50 per share** (6-month return = 50/150 = **+33.33%**).

b. Bearish Speculation in Stock (Political Condition Worsens):
• **Strategy: Short-Sell MBL Stock (Short Stock Position)** at S_0 = Rs 150 per share (or avoid buying; if already bought Long at Rs 150, a drop to Rs 100 causes a loss of **-Rs 50 per share**; if Short-Sold at Rs 150 and bought back at Rs 100, gain is **+Rs 50 per share**).

c. Bullish Speculation in Options (Political Condition Settles Soon):
• **Strategy: Buy Call Options (Long Call)** at X = Rs 150 for premium C = Rs 10 per share.
• **Gain at S_T = Rs 200:** Profit = max(200 - 150, 0) - 10 = 50 - 10 = **+Rs 40 per call option** (6-month return = 40/10 = **+400%**; and if Rs 150 is invested in 15 calls, total profit = 15 x Rs 40 = **+Rs 600**).
• **Breakeven Stock Price (BEP):** X + C = Rs 150 + Rs 10 = **Rs 160 per share**.

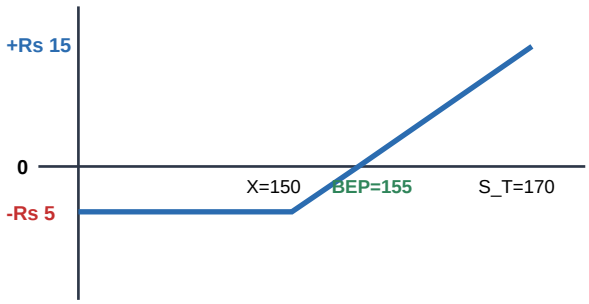
d. Bearish Speculation in Options (Political Condition Worsens):
• **Strategy: Buy Put Options (Long Put)** at X = Rs 150 for premium P = Rs 5 per share.
• **Gain at S_T = Rs 100:** Profit = max(150 - 100, 0) - 5 = 50 - 5 = **+Rs 45 per put option** (6-month return = 45/5 = **+900%**; and if Rs 150 is invested in 30 puts, total profit = 30 x Rs 45 = **+Rs 1,350**).
• **Breakeven Stock Price (BEP):** X - P = Rs 150 - Rs 5 = **Rs 145 per share**.

e. Bullish Option Strategy Without Using Call Options:
• **Strategy: Sell (Write) Put Options (Short Put)** at X = Rs 150 and collect premium P = Rs 5 per share (or create a *Synthetic Long Call = Buy Stock @ Rs 150 + Buy Put @ Rs 5 = Rs 155*).
- **Under Short Put:** At S_T = Rs 200, the put expires worthless and your profit is the premium collected = **+Rs 5 per put**; Breakeven price = X - P = 150 - 5 = **Rs 145**.
- **Under Synthetic Call (Long Stock + Long Put):** At S_T = Rs 200, profit = (200 - 150) - 5 = **+Rs 45 per share**; Breakeven price = S_0 + P = 150 + 5 = **Rs 155**.

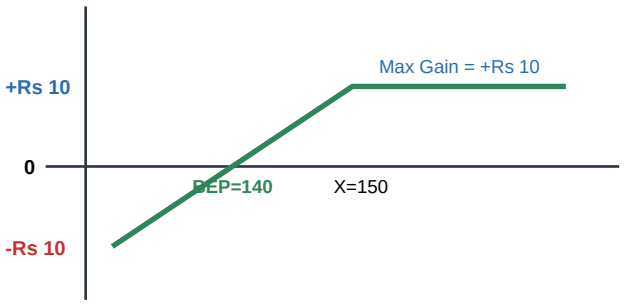
f & g. Hedging Long Stock Position with Put Option [Protective Put: Long Stock + Long Put] (Part f) and with Call Option [Covered Call: Long Stock + Short Call] (Part g):
• **Part (f) Strategy: Protective Put** (Hold 1 Share bought at Rs 150 + Buy 1 Put at X = Rs 150 for P = Rs 5 => Total Cost = Rs 155).
- **Breakeven Stock Price (Part f):** S_0 + P = Rs 150 + Rs 5 = **Rs 155 per share**.
• **Part (g) Strategy: Covered Call** (Hold 1 Share bought at Rs 150 + Sell/Write 1 Call at X = Rs 150 for C = +Rs 10 => Net Cost = Rs 140).
- **Breakeven Stock Price (Part g):** S_0 - C = Rs 150 - Rs 10 = **Rs 140 per share**.

Expiration Stock Price (S_T)	Stock Gain/Loss (S_T - 150)	Part (f) Long Put Profit max(150-S_T,0) - 5	Part (f) Protective Put Total Profit/Loss	Part (g) Short Call Profit 10 - max(S_T-150,0)	Part (g) Covered Call Total Profit/Loss
Rs 130	-Rs 20	+Rs 15	**-Rs 5 (Max Loss)**	+Rs 10	-Rs 10 (Loss)
Rs 140	-Rs 10	+Rs 5	**-Rs 5 (Max Loss)**	+Rs 10	**Rs 0 (Part g BEP)**
Rs 150	Rs 0	-Rs 5	**-Rs 5 (Max Loss)**	+Rs 10	**+Rs 10 (Max Gain)**
Rs 155 (Part f BEP)	+Rs 5	-Rs 5	**Rs 0 (Part f BEP)**	+Rs 5	**+Rs 10 (Max Gain)**
Rs 160	+Rs 10	-Rs 5	**+Rs 5 (Profit)**	Rs 0	**+Rs 10 (Max Gain)**
Rs 170	+Rs 20	-Rs 5	**+Rs 15 (Profit)**	-Rs 10	**+Rs 10 (Max Gain)**

(f) Protective Put Profit Graph (Long Stock + Long Put)



(g) Covered Call Profit Graph (Long Stock + Short Call)



EXAM HINT TO REMEMBER: *Compare (f) Protective Put vs. (g) Covered Call on a Rs 150 stock: Protective Put (Long Stock + Long Put @ Rs 5) caps downside loss at -Rs 5 with BEP = 150 + 5 = Rs 155 and unlimited upside. Covered Call (Long Stock + Short Call @ Rs 10) cushions downside by Rs 10 with BEP = 150 - 10 = Rs 140 and caps upside profit at +Rs 10!*

PAPER 2: SHANKER DEV CAMPUS (TU) — BNK 221: CORPORATE FINANCING DECISION

BBA / Eighth Semester | Preboard Examination 2026 | Full Marks: 100 | Time: 3 Hrs

Group A: Brief Answer Questions [10 x 2 = 20]

Q1. What do you mean by a corporate financing decision?

Answer: Corporate financing decision refers to the strategic managerial decision regarding how a firm raises short-term and long-term funds from internal and external sources (debt, preferred stock, and common equity) to finance its assets and operations so as to minimize its weighted average cost of capital (WACC) and maximize firm value.

Q2. Define optimal capital structure.

Answer: Optimal capital structure is the best proportion of long-term debt, preferred stock, and common equity financing that balances the tax-shield benefits of debt against the costs of financial distress, thereby **minimizing the firm's overall weighted average cost of capital (WACC)** and **maximizing the market value of the firm** (and share price).

Q3. What is right offering?

Answer: A rights offering (or privileged subscription) is a primary-market method of raising new common equity in which a corporation grants its existing common shareholders the first right (preemptive right) to purchase a pro-rata number of newly issued shares at a discounted subscription price (P_s) within a specified period.

Q4. Gorkha Herbal company purchase under 2/10 net 30. Annual purchase of the company is Rs. 100,000. What is the annual percentage cost of not taking discount? Also calculate the effective annual rate. Assume 360 days in a year.

Answer (Step-by-Step Numerical Solution):
Given: Credit terms = 2/10, net 30 => d = 2% = 0.02; DP = 10 days; CP = 30 days; Days in year = 360.
• **1. Annual Percentage Cost (Nominal Rate, APC):**
APC = [d / (1 - d)] x [360 / (CP - DP)]
APC = (0.02 / 0.98) x [360 / (30 - 10)] = 0.020408 x (360 / 20) = 0.020408 x 18 = **36.73%**
• **2. Effective Annual Rate (Compounded Rate, EAR):**
EAR = [1 + d / (1 - d)]^[360 / (CP - DP)] - 1
EAR = (1 + 0.02 / 0.98)^18 - 1 = (1.020408)^18 - 1 = 1.438569 - 1 = **43.86%**

EXAM HINT TO REMEMBER: *For 2/10 net 30 (360-day year): Number of 20-day credit periods per year = 360 / 20 = 18 periods, and periodic rate = 2/98 = 2.0408%. Nominal APC = 18 x 2.0408% = 36.73%; Compounded EAR = (1.020408)^18 - 1 = 43.86%!*

Q5. B and B Cafe are planning to borrow Rs. 1,000,000 for one year from Nepal Bank. The bank has quoted 10 percent interest but requires 10 percent compensating balance. What is the effective cost of bank loan?

Answer (Step-by-Step Numerical Solution):
Given: Loan amount = Rs 1,000,000; Stated annual interest rate (i) = 10% = 0.10; Compensating balance ratio (CB) = 10% = 0.10.
Formula for Effective Annual Cost of Simple Interest Loan with Compensating Balance:
Effective Cost = i / (1 - CB) = 0.10 / (1 - 0.10) = 0.10 / 0.90 = 0.111111 = **11.11%**
*(Verification: Annual interest = Rs 100,000; Usable funds = Rs 1,000,000 - Rs 100,000 = Rs 900,000; Effective rate = 100,000 / 900,000 = **11.11%**.)*

EXAM HINT TO REMEMBER: *Simple loan with Compensating Balance: Effective Rate = i / (1 - CB) = 10% / (1 - 0.10) = 10/90 = 11.11%.*

Q6. Tara Pandey, manager of Star Trading is planning to purchase a car. Market price of the car is Rs. 5,000,000. NB bank provides car loan at 10 percent interest rate. Loan period is 5 years. Calculate equal monthly installment (EMI) amount.

Answer (Step-by-Step Numerical Solution):
Given:
• Loan principal (P_0) = Rs 5,000,000
• Annual interest rate = 10% p.a. => **Monthly interest rate (r_m) = 10% / 12 = 0.0083333 (0.8333% per month)**
• Loan period = 5 years => **Total monthly installments (n) = 5 x 12 = 60 months**

Step 1: Present Value Interest Factor of Annuity for 60 Months at 0.8333% per Month:
PVIFA(0.8333%, 60) = [1 - (1 + 0.10 / 12)^(-60)] / (0.10 / 12) = (1 - 0.607789) / 0.0083333 = **47.065369**

Step 2: Equal Monthly Installment (EMI):
EMI = Loan Principal / PVIFA(0.8333%, 60) = Rs 5,000,000 / 47.065369 = **Rs 106,235.22 per month**
(Note: If calculated on an annual installment basis first: PVIFA(10%, 5 yrs) = 3.790787 -> Annual PMT = Rs 1,318,987.40 -> divided by 12 = Rs 109,915.62.)

EXAM HINT TO REMEMBER: *For Monthly EMI on a 5-year 10% loan: always convert years to months (n = 5 x 12 = 60) and annual rate to monthly rate (r_m = 0.10 / 12 = 0.008333). Then EMI = Loan / PVIFA(r_m, 60) = Rs 106,235.22!*

Q7. A Rs 1000 face value convertible debenture is convertible into 10 shares of common stock. The market price per share is Rs 80. What is the conversion price? The conversion premium in percent?

Answer (Step-by-Step Numerical Solution):
Given: Face value (Par) = Rs 1,000; Conversion Ratio (CR) = 10 shares; Current stock price (P_0) = Rs 80.
• **1. Conversion Price (P_c):**
P_c = Face Value / Conversion Ratio = Rs 1,000 / 10 shares = **Rs 100 per share**
• **2. Conversion Premium in Percent:**
Conversion Premium (%) = [(P_c - P_0) / P_0] x 100 = [(100 - 80) / 80] x 100 = (20 / 80) x 100 = **25%**
*(Equivalently, using Conversion Value = 10 x Rs 80 = Rs 800: Premium = [(1,000 - 800) / 800] x 100 = **25%**.)*

EXAM HINT TO REMEMBER: *Conversion Price = Par Value / Conversion Ratio = 1,000 / 10 = Rs 100. Conversion Premium (%) = (Conversion Price - Market Price) / Market Price = (100 - 80) / 80 = 25%.*

Q8. What is a warrant? List out its features.

Answer: A **warrant** is a long-term option issued by a corporation that gives the holder the right to buy a stated number of shares of the company's common stock at a specified exercise price within a stated time period.
Key Features of Warrants:
• **1. Fixed Exercise Price & Exercise Ratio:** Specifies the exact price per share and number of shares each warrant can buy.
• **2. Expiration Date:** Has a stated long-term maturity (typically 3 to 10 years, or occasionally perpetual).
• **3. Detachability:** Most warrants are detachable from the host bond and can be traded separately on the stock exchange.
• **4. No Voting or Dividend Rights:** Warrant holders receive no dividends and have no voting power until they exercise the warrants.

Q9. Write the meaning of preferred stock.

Answer: Preferred stock (preference share) is a hybrid long-term corporate security that combines features of both debt and common equity: it pays a fixed periodic dividend that must be paid before any dividend can be paid to common shareholders, and holds a senior priority claim over common stock on assets in liquidation, but typically carries no voting rights.

Q10. What is meant by bankruptcy liquidation?

Answer: Bankruptcy liquidation is the formal court-supervised legal process of terminating an insolvent corporation's operations (when the firm is "worth more dead than alive"), appointing a trustee/liquidator to sell off all the company's assets for cash, and distributing the proceeds to claimants according to the strict statutory absolute priority rule.

Group B: Short Answer Questions (Attempt any SIX — ALL Solved) [6 x 5 = 30]

Q11. What do you mean by capital structure? What are the factors affecting capital structure decision of a firm? Explain.

Answer:
Meaning of Capital Structure:
Capital structure refers to the permanent long-term financing mix of a corporation, represented by the relative proportions of long-term debt, preferred stock, and common shareholders' equity (common stock plus retained earnings) used to finance its overall operations and capital investments.

Major Factors Affecting the Capital Structure Decision of a Firm:
• **1. Business Risk and Sales Stability:** Firms with stable, predictable revenues and cash flows can safely assume higher fixed debt obligations than volatile, cyclical businesses.
• **2. Asset Structure (Tangibility):** Companies possessing general-purpose tangible fixed assets (land, buildings, equipment) can pledge collateral to borrow at lower interest rates.
• **3. Operating Leverage:** Firms with high fixed operating costs already face high operating risk and must limit financial leverage (debt) to keep total risk in check.
• **4. Corporate Tax Rate:** Because interest expense is tax-deductible, a higher corporate tax rate increases the value of the debt tax shield.

• **5. Profitability and Internal Cash Flow:** Highly profitable firms often finance investments internally through retained earnings (following the Pecking Order Theory) rather than issuing external debt or equity.
• **6. Control Considerations & Managerial Conservatism:** If existing owners fear dilution of voting control, they prefer debt over issuing new common stock; conversely, risk-averse managers use less debt.
• **7. Growth Rate and Financial Flexibility:** Fast-growing firms need to maintain reserve borrowing capacity to fund future investment opportunities.

Q12. Define venture capital. Explain the importance of venture capital financing.

Answer:

Definition of Venture Capital:

Venture capital (VC) is institutional private equity capital provided by specialized venture capital firms to early-stage, innovative, and high-growth-potential unlisted companies that carry high risk and lack tangible collateral for traditional bank loans.

Importance of Venture Capital Financing:

• **1. Bridging the Startup Funding Gap:** Provides critical risk capital to innovative technology, healthcare, agribusiness, and service startups that cannot access bank loans or public IPO markets.
• **2. Active Mentorship and Strategic Value Addition:** VCs contribute board governance, strategic guidance, industry contacts, and executive recruitment support.
• **3. Risk Sharing Without Fixed Interest Burden:** Because VC is invested as equity or convertible preferred stock, young firms are not burdened by monthly debt service during their early development stage.
• **4. Credibility and IPO Readiness:** Prepares startups with strong financial controls and corporate governance to eventually go public (IPO) or achieve a profitable strategic acquisition.
• **5. Fostering Innovation, Entrepreneurship, and Job Creation:** Accelerates technological commercialization and skilled employment across the economy.

Q13. How a solvent company can be liquidated in Nepal? Explain.

Answer:

Under the **Companies Act, 2063 (2006) of Nepal (Chapter 10: Voluntary Liquidation of a Company)**, a **solvent company**—one that is fully capable of paying all its debts and liabilities in full—can be liquidated voluntarily by its shareholders through the following legal procedure:

• **Step 1: Solvency Declaration by the Board of Directors:**

Before calling a general meeting, the Board of Directors must make a formal written declaration (accompanied by an audited financial statement of assets and liabilities) confirming that they have thoroughly examined the company's affairs and that the company has no debts or is able to pay all its debts and liabilities in full within one year from the commencement of liquidation.

• **Step 2: Adoption of Special Resolution in the General Meeting:**

A copy of the directors' declaration and audited accounts is presented to shareholders, and the General Meeting passes a **Special Resolution** for voluntary liquidation (requiring at least 75% majority vote).

• **Step 3: Appointment of a Licensed Liquidator and Auditor:**

In the same general meeting, the shareholders appoint a licensed Insolvency Practitioner / Liquidator and an auditor, fixing their remuneration, and notify the **Office of the Company Registrar (OCR)** within 7 days.

• **Step 4: Cessation of Business Operations & Takeover by Liquidator:**

Upon appointment of the liquidator, the powers of the Board of Directors cease, and the liquidator takes custody and control of all assets, accounts, and properties of the company.

• **Step 5: Realization of Assets and Settlement of All Claims:**

The liquidator publishes a notice inviting any creditor claims, sells/realizes the company's assets, pays liquidation expenses, settles all employee dues, taxes, and creditors in full, and distributes the remaining surplus capital among preferred and common shareholders according to the Articles of Association.

• **Step 6: Final Audit Report, OCR Cancellation of Registration & Dissolution:**

The liquidator submits the final audited report of receipts and payments to the Office of the Company Registrar (OCR). The OCR strikes the company's name off the company register and publishes a dissolution notice, completing the liquidation.

Q14. Yonge Company must arrange financing for its working capital requirements for the coming year. Yonge can (a) borrow from its bank on a simple interest basis (interest payable at the end of the loan) for 1 year at a 12 percent nominal rate; (b) borrow on a 3-month, but renewable, loan at an 11.5 percent nominal rate; (c)

borrow on an installment loan basis at a 6 percent add-on rate with 12 end-of-month payments; or (d) obtain the needed funds by no longer taking discounts and thus increasing its accounts payable. Yonge buys on terms of 1/15, net 60. What is the effective annual cost (not the nominal cost) of the least expensive type of credit, assuming 365 days per year?

Answer (Step-by-Step Numerical Solution):
Given: 4 short-term financing alternatives; 365 days in a year. Let us compute the **Effective Annual Rate (EAR)** of each alternative:

Alternative (a): 1-Year Simple Interest Bank Loan at 12% Nominal Rate:
Since interest is paid once at the end of 1 year (m = 1):
EAR_(a) = 12.00%

Alternative (b): 3-Month Renewable Loan at 11.5% Nominal Rate:
• Quarterly compounding (m = 4 quarters per year):
EAR_(b) = (1 + 0.115 / 4)^4 - 1 = (1.02875)^4 - 1 = 1.120055 - 1 = **12.01%**
*(Using 365/90 = 4.0556 periods: EAR = [1 + 0.115 x (90/365)]^(365/90) - 1 = **12.01%**)*

Alternative (c): 1-Year Installment Loan at 6% Add-On Rate with 12 End-of-Month Payments:
• For every Re 1 borrowed, total repayable = 1.06; Monthly payment = 1.06 / 12 = 0.088333
• **Exact TVM / IRR Method:**
PVIFA(r_m, 12) = 1 / 0.088333 = 11.320755 => Monthly rate r_m = **0.9080% per month** (APR = 10.90%)
Exact EAR_(c) = (1 + 0.009080)^12 - 1 = **11.46%**
• **Approximate Add-On Formula:**
Approximate APR = (2 x m x I) / [P x (m + 1)] = (2 x 12 x 0.06) / (1 x 13) = 1.44 / 13 = **11.08%**
Compounded EAR_(c) = (1 + 0.110769 / 12)^12 - 1 = **11.66%**

Alternative (d): Foregoing Trade Credit Discounts on Terms of 1/15, Net 60 (365 Days):
• Periodic cost rate = d / (1 - d) = 0.01 / (1 - 0.01) = 0.01 / 0.99 = **0.010101 (1.0101%)**
• Number of 45-day credit periods in a 365-day year = 365 / (60 - 15) = 365 / 45 = **8.1111 periods**
(Nominal Cost APC = 0.010101 x 8.1111 = 8.19%)
• **Effective Annual Rate (EAR_(d)):**
EAR_(d) = (1 + 0.01 / 0.99)^(365 / 45) - 1 = (1.010101)^(8.1111) - 1 = 1.084934 - 1 = **8.49%**

Conclusion / Least Expensive Credit:
Comparing all four effective annual rates: (a) **12.00%**, (b) **12.01%**, (c) **11.46% (or 11.66%)**, and (d) **8.49%**, the least expensive type of credit is **Alternative (d) (Trade Credit / Stretching Accounts Payable on 1/15, net 60)** with an Effective Annual Cost (EAR) of **8.49%**.

EXAM HINT TO REMEMBER: *Usually trade credit is the most expensive, but when the discount is tiny (only 1%) and the credit period is long (60 - 15 = 45 days), trade credit becomes the CHEAPEST option at 8.49% EAR! Always compute all 4 EARs before deciding!*

Q15. The Boat Company has current earnings of Rs 3 a share with 500,000 shares outstanding. The company plans to issue 40,000 shares of 7 percent, Rs 50 par value convertible preferred stock at par. The preferred stock is convertible into two shares of common for each preferred share held. The common stock has a current market price of Rs 21 per share. a. What is the preferred stock's conversion value? b. What is its conversion premium? c. If total earnings stay the same, what will be the effect of the issue on basic earnings per share (1) before conversion? (2) on a fully diluted basis?

Answer (Step-by-Step Numerical Solution):
Given:
• Current EPS = Rs 3.00 per share; Existing common shares (N_0) = 500,000 shares
=> **Current Total Earnings (Net Income) = Rs 3.00 x 500,000 = Rs 1,500,000**
• Convertible Preferred Stock Issue: 40,000 shares of 7%, Rs 50 par value issued at par (Rs 50)
=> Total Preferred Dividend = 40,000 shares x Rs 50 x 7% = 40,000 x Rs 3.50 = **Rs 140,000 per year**
• Conversion Ratio (CR) = 2 shares of common stock per preferred share (Total new common shares = 40,000 x 2 = **80,000 shares**)
• Current market price of common stock (P_0) = Rs 21 per share

a. Preferred Stock's Conversion Value:
Conversion Value = Conversion Ratio (CR) x Current Market Price of Common Stock (P_0)
Conversion Value = 2 shares x Rs 21 = **Rs 42 per preferred share**

b. Conversion Premium (in Rupees and Percentage):
• **In Rupees per Preferred Share:** Issue Price (Par) - Conversion Value = Rs 50 - Rs 42 = **Rs 8 per preferred share**

*(Or per Common Share: Conversion Price = Rs 50 / 2 = Rs 25; Premium = Rs 25 - Rs 21 = **Rs 4 per common share**)*
• **In Percentage:** (Rs 8 / Rs 42) x 100 = **19.05%**

c. Effect on Basic and Fully Diluted Earnings Per Share (Assuming Total Earnings Stay at Rs 1,500,000):
• **(1) Basic EPS BEFORE Conversion:**
- Earnings available to common stockholders = Total Earnings - Preferred Dividends = Rs 1,500,000 - Rs 140,000 = Rs 1,360,000
- **Basic EPS (Before Conversion)** = Rs 1,360,000 / 500,000 shares = **Rs 2.72 per share** (decreases from Rs 3.00 to Rs 2.72).
• **(2) Diluted EPS on a FULLY DILUTED Basis (After Conversion):**
- Upon full conversion, preferred dividends (Rs 140,000) are no longer paid, so all Rs 1,500,000 is available to common shares, and common shares increase to 500,000 + (40,000 x 2) = **580,000 shares**:
- **Fully Diluted EPS (After Conversion)** = Rs 1,500,000 / 580,000 shares = **Rs 2.5862 (or Rs 2.59 per share)**.

***EXAM HINT TO REMEMBER:** For Convertible Preferred Stock in Q15(c): (1) Before conversion, subtract Preferred Dividend (40,000 x Rs 3.50 = Rs 140,000) from Rs 1,500,000 and divide by 500,000 shares -> Rs 2.72. (2) After conversion, DO NOT subtract Preferred Dividend—divide the full Rs 1,500,000 by (500,000 + 80,000 = 580,000 shares) -> Rs 2.59!*

Q16. The Public Service (PS) company has warrants outstanding that allow the holder to purchase 3 shares of stock for a total of Rs. 60 for each warrant. Currently, the market price per share of the company common stock is Rs. 18. Investors hold the following probabilistic beliefs about the stock 6 months hence: [Market price per share: Rs 16 (Prob 0.15), Rs 18 (Prob 0.20), Rs 20 (Prob 0.30), Rs 22 (Prob 0.20), Rs 24 (Prob 0.15)] a. What is the present theoretical value of the warrant? b. What is the expected value of stock price 6 months hence? c. What is the theoretical value of the warrant 6 months hence?

Answer (Step-by-Step Numerical Solution):
Given:
• Shares purchased per warrant (N) = 3 shares; Total exercise cost per warrant = Rs 60
=> **Exercise price per share (E) = Rs 60 / 3 = Rs 20 per share**
• Current market price per share (P_0) = Rs 18

a. Present Theoretical Value of the Warrant:
• Formula TV = (P_0 - E) x N = (18 - 20) x 3 = (18 x 3) - 60 = 54 - 60 = **-Rs 6**
• Since a warrant cannot sell for a negative price, its effective **Present Theoretical Value (Floor Value) = Rs 0**.

b. Expected Value of Stock Price 6 Months Hence, E(P):
E(P) = (16 x 0.15) + (18 x 0.20) + (20 x 0.30) + (22 x 0.20) + (24 x 0.15)
E(P) = 2.40 + 3.60 + 6.00 + 4.40 + 3.60 = **Rs 20.00 per share**

c. Expected Theoretical Value of the Warrant 6 Months Hence:
• **Method 1 (Evaluated at Expected Stock Price E(P) = Rs 20.00):**
TV = (E(P) - E) x N = (20 - 20) x 3 = **Rs 0**
• **Method 2 (Probability-Weighted Expected Warrant Value Across All 5 Stock Price States):**
- At P = Rs 16, Rs 18, Rs 20: Warrant is out-of-the-money / at-the-money -> TV = **Rs 0**
- At P = Rs 22 (Prob 0.20): TV = (22 - 20) x 3 = **Rs 6**
- At P = Rs 24 (Prob 0.15): TV = (24 - 20) x 3 = **Rs 12**
- **Expected Theoretical Value E(TV)** = (0 x 0.65) + (6 x 0.20) + (12 x 0.15) = 0 + 1.20 + 1.80 = **Rs 3.00**

***EXAM HINT TO REMEMBER:** In Q16(c), show BOTH Method 1 [TV at E(P) = (20 - 20) x 3 = Rs 0] and Method 2 [Expected TV across states = 6(0.20) + 12(0.15) = Rs 3.00] to guarantee full marks!*

Q17. As one of the minority shareholders of the Bishal Trading Corporation, you are dissatisfied with current operations of the company. You feel that if you could gain membership on the company's board of directors, you could pursue the company to make improvements. The problem is that current management controls 70 percent of the stock, you control only 10 percent and the balance is held by the other minority shareholders. There are a total of 500,000 voting shares. Ten directors will be elected at the next annual stockholders meeting. a. If voting is non-cumulative, can you elect yourself director? b. Suppose you can persuade all the minority shareholders that you should be elected. If voting is non-cumulative, can they select you? c. If voting is cumulative, can you elect yourself as a director? d. What percent of minority shares other than you own will you need to have voted for you to be certain of election? e. What is the number of directors the minority shareholders can elect with certainty?

Answer (Step-by-Step Numerical Solution):
Given:

• Total voting shares (N) = 500,000 shares; Total directors to be elected (D) = 10 directors
• Ownership breakdown:
- Current Management (70%) = 0.70 x 500,000 = **350,000 shares**
- Your Own Shares (10%) = 0.10 x 500,000 = **50,000 shares**
- Other Minority Shareholders (20%) = 0.20 x 500,000 = **100,000 shares**
- Total Minority Shares (You + Other Minority = 30%) = **150,000 shares**

a. If Voting is Non-Cumulative (Majority Rule), Can You Elect Yourself?
No. Under non-cumulative (majority) voting, you need 50% + 1 = **250,001 shares** to elect any director. Since you only control 10% (50,000 shares) and management controls 70% (350,000 shares), management will elect all 10 directors.

b. If Voting is Non-Cumulative and All Minority Shareholders Support You, Can They Elect You?
No. Even if all minority shareholders combine their votes (10% + 20% = **30% or 150,000 shares**), they still have less than the required majority of 250,001 shares (and management holds 70% = 350,000 shares).

c. If Voting is Cumulative, Can You Elect Yourself as a Director?
• Minimum shares required to guarantee election of 1 director (d = 1) under cumulative voting:
Min Shares = [(N x d) / (D + 1)] + 1 = [(500,000 x 1) / (10 + 1)] + 1 = 45,454.55 + 1 = **45,455 shares** (or 9.09% + 1 share).
• Since you already own 10% = **50,000 shares** (which is greater than 45,455 shares), **YES, you can elect yourself as a director on your own!**

d. What Percent of Minority Shares Other Than Your Own Will You Need for Certain Election?
Because your own 50,000 shares (10%) already exceed the 45,455 shares required to guarantee 1 seat under cumulative voting, you need **0% (none) of the other minority shareholders' shares**!

e. Number of Directors (d) That All Minority Shareholders Together (30% = 150,000 Shares) Can Elect with Certainty:
• Using the cumulative voting formula:
[(500,000 x d) / (10 + 1)] + 1 <= 150,000
(500,000 x d) / 11 <= 149,999 => d <= (149,999 x 11) / 500,000 = **3.2999 directors**
• Rounding down to the nearest whole integer, all minority shareholders together can elect **3 directors** with certainty (which requires [(500,000 x 3)/11] + 1 = **136,364 shares**).

EXAM HINT TO REMEMBER: *With 10 directors, 1 seat under cumulative voting requires only 1/(10+1) = 1/11 = 9.09% + 1 share (45,455 shares). Since you already own 10% (50,000 shares), you can elect yourself SOLO (needing 0% from other minority holders), and all 30% minority together can elect floor(30% / 9.09%) = 3 directors!*

Group C: Long Answer Questions (Attempt any THREE — ALL Solved) [3 x 10 = 30]

Q18. Ram Pandit total assets consist of 490 shares Bright Land corporation and Rs 200,000 in cash. Bright Land now offers stockholders one additional share at a price of Rs 100 for each five shares held. The current market price of stock is Rs 160. a. What is the value of each right? b. What is the Mr. Pandit's Total Assets before right offerings? c. What is the total assets after right offering for each of these alternative: a. He exercises all his rights. b. He sells all his rights. c. He sells 400 rights and exercise 90 rights. d. He neither sells nor exercises the rights.

Answer (Step-by-Step Numerical Solution):
Given:
• Ram Pandit's holdings = 490 shares + Rs 200,000 cash
• Rights required per new share (#) = 5 rights
• Subscription price (P_s) = Rs 100; Rights-on stock price (P_0) = Rs 160

a. Theoretical Value of Each Right (V_r) and Ex-Rights Stock Price (P_e):
• **Value of One Right (V_r):**
V_r = (P_0 - P_s) / (# + 1) = (160 - 100) / (5 + 1) = 60 / 6 = **Rs 10 per right**
• **Ex-Rights Price of Stock (P_e):**
P_e = P_0 - V_r = Rs 160 - Rs 10 = **Rs 150 per share**

b. Mr. Pandit's Total Assets BEFORE Rights Offering:
• Value of 490 shares @ Rs 160 = 490 x Rs 160 = Rs 78,400
• Cash balance = Rs 200,000
• **Total Assets Before Rights Offering = Rs 78,400 + Rs 200,000 = Rs 278,400**

c. Mr. Pandit's Total Assets AFTER Rights Offering Under All 4 Alternatives:
• **Alternative (a): He Exercises All His 490 Rights:**
- New shares purchased = 490 rights / 5 rights per share = **98 new shares**

- Cash paid for 98 new shares @ Rs 100 = 98 x Rs 100 = Rs 9,800
- Value of Total Shares (490 + 98 = 588 shares) @ Ex-Rights Price (Rs 150) = 588 x Rs 150 = Rs 88,200
- Remaining Cash Balance = Rs 200,000 - Rs 9,800 = Rs 190,200
- **Total Assets = Rs 88,200 + Rs 190,200 = Rs 278,400**

• **Alternative (b): He Sells All His 490 Rights (@ Rs 10 each):**

- Value of 490 shares @ Ex-Rights Price (Rs 150) = 490 x Rs 150 = Rs 73,500
- Cash from selling 490 rights = 490 x Rs 10 = Rs 4,900
- New Cash Balance = Rs 200,000 + Rs 4,900 = Rs 204,900
- **Total Assets = Rs 73,500 + Rs 204,900 = Rs 278,400**

• **Alternative (c): He Sells 400 Rights and Exercises 90 Rights:**

- New shares bought with 90 rights = 90 / 5 = **18 new shares** (cost = 18 x Rs 100 = -Rs 1,800)
- Cash received from selling 400 rights = 400 x Rs 10 = +Rs 4,000
- Value of Total Shares (490 + 18 = 508 shares) @ Ex-Rights Price (Rs 150) = 508 x Rs 150 = Rs 76,200
- New Cash Balance = Rs 200,000 + Rs 4,000 - Rs 1,800 = Rs 202,200
- **Total Assets = Rs 76,200 + Rs 202,200 = Rs 278,400**

• **Alternative (d): He Neither Sells Nor Exercises the Rights (Lets Rights Expire Worthless):**

- Value of 490 shares @ Ex-Rights Price (Rs 150) = 490 x Rs 150 = Rs 73,500
- Cash Balance (unchanged) = Rs 200,000
- **Total Assets = Rs 73,500 + Rs 200,000 = Rs 273,500 (Suffers a wealth loss of Rs 4,900!)**

EXAM HINT TO REMEMBER: *Notice how Alternatives (a), (b), and (c) all preserve Mr. Pandit's exact wealth of Rs 278,400, whereas Alternative (d) (doing nothing) loses the value of 490 rights = 490 x Rs 10 = Rs 4,900, dropping his wealth to Rs 273,500!*

Q19. South-West Agro Company is facing liquidity crises since few months. It is planning to loan Rs 1,000,000 for 30 days. Having no source of additional unsecured borrowing, the firm must find a secured short-term lender. The firm is considering using inventories or accounts receivable as collateral. The value of inventory is Rs 3,000,000. The company has the following alternatives: a. Local Bank will make a Rs 1,000,000 trust receipt loan against the finished goods inventory. The annual interest rate on the loan is 10 percent on the outstanding loan balance plus a 0.10 percent administration fee levied against the Rs 1,000,000 initial loan amount. Because it will be liquidated as inventory is sold, the average amount owed over the month is expected to be Rs 750,000. b. Regional Bank is willing to lend Rs 1,000,000 against a floating lien on the value of inventory for 30-day period at an annual interest rate of 11 percent. c. National Bank will loan Rs 1,000,000 against the finished goods inventory and charge 13 percent annual interest on the outstanding loan balance. A 0.1 percent warehousing fee will be levied against the average amount borrowed. Because the loan will be liquidated as inventory is sold, the average loan balance is expected to be Rs 700,000. d. Suppose the company is considering to raise funds through factoring arrangement. The average receivables outstanding are Rs 1,500,000 and terms are net 30 days. Bishal Bank offers factoring for a 3 percent factoring commission charging 1.2 percent interest per month on advances and requiring a 20 percent factor's reserve. Under this plan, the firm would factor all accounts and close its credit and collection department, saving Rs 480,000 per year. Calculate the cost of each of the proposed plans for obtaining an initial loan amount of Rs 1,000,000. Which plan should the company use? (2+1+2+4+1)

Answer (Step-by-Step Numerical Solution):

(Calculations are shown using a 360-day year [30/360 = 1/12 year], with 365-day year figures also noted.)

a. Plan A — Local Bank (Trust Receipt Inventory Loan) [2 Marks]:

• 30-Day Interest on Average Balance (Rs 750,000) @ 10% p.a.:
Interest = Rs 750,000 x 0.10 x (30 / 360) = **Rs 6,250.00** *(Using 365d: Rs 6,164.38)*
• Administration Fee (0.10% on Rs 1,000,000 initial loan):
Fee = Rs 1,000,000 x 0.0010 = **Rs 1,000.00**
• **Total 30-Day Rupee Cost (Plan A)** = Rs 6,250 + Rs 1,000 = **Rs 7,250.00**
• **30-Day Periodic Cost Rate** (on Rs 750,000 avg loan) = 7,250 / 750,000 = **0.9667%** => **Annualized Cost (APR) = 0.9667% x 12 = 11.60%** (or on Rs 1,000,000 initial loan: 7,250 / 1,000,000 x 12 = **8.70%**).

b. Plan B — Regional Bank (Floating Lien Inventory Loan) [1 Mark]:

• 30-Day Interest on Rs 1,000,000 @ 11% p.a.:
Total 30-Day Rupee Cost (Plan B) = Rs 1,000,000 x 0.11 x (30 / 360) = **Rs 9,166.67** *(Using 365d: Rs 9,041.10)*

• **Annualized Cost (APR) = 11.00%** (Compounded EAR = (1 + 0.11/12)^12 - 1 = 11.57%).

c. Plan C — National Bank (Warehouse Receipt Inventory Loan) [2 Marks]:
• 30-Day Interest on Average Balance (Rs 700,000) @ 13% p.a.:
Interest = Rs 700,000 x 0.13 x (30 / 360) = **Rs 7,583.33** *(Using 365d: Rs 7,479.45)*
• Warehousing Fee (0.10% on Rs 700,000 average balance):
Fee = Rs 700,000 x 0.0010 = **Rs 700.00**
• **Total 30-Day Rupee Cost (Plan C)** = Rs 7,583.33 + Rs 700.00 = **Rs 8,283.33**
• **30-Day Periodic Cost Rate** (on Rs 700,000 avg loan) = 8,283.33 / 700,000 = **1.1833%** => **Annualized Cost (APR) = 1.1833% x 12 = 14.20%** (or on Rs 1,000,000 initial loan: 8,283.33 / 1,000,000 x 12 = **9.94%**).

d. Plan D — Bishal Bank (Factoring Accounts Receivable) [4 Marks]:
• **Maximum Available Advance Check:**
- Monthly Accounts Receivable = Rs 1,500,000
- Less: Factor's Reserve (20% of Rs 1,500,000) = -Rs 300,000
- Less: Factoring Commission (3% of Rs 1,500,000) = -Rs 45,000
- Gross Funds Available Before Interest = **Rs 1,155,000** (sufficient to provide the required Rs 1,000,000 loan!).
• **Monthly Cost of Factoring (for Obtaining Rs 1,000,000 Advance):**
- Factoring Commission (3% of Rs 1,500,000) = Rs 45,000
- Monthly Interest on Rs 1,000,000 Advance @ 1.2% per month = Rs 1,000,000 x 0.012 = Rs 12,000
(Note: If maximum advance is taken, interest on Rs 1,155,000 @ 1.2% = Rs 13,860 -> Net usable advance = Rs 1,141,140)
- Total Gross Monthly Factoring Charges = Rs 45,000 + Rs 12,000 = Rs 57,000
- Less: Monthly Credit & Collection Department Savings (Rs 480,000 / 12 months) = -Rs 40,000
- **Net 30-Day Rupee Cost of Factoring (Plan D)** = Rs 57,000 - Rs 40,000 = **Rs 17,000.00**
(Or Rs 18,860 if maximum advance is taken)
- **Annualized Percentage Cost (Plan D)** = (Rs 17,000 / Rs 1,000,000) x 12 = **20.40% p.a.** (or on max advance: (18,860 / 1,141,140) x 12 = **19.83% p.a.**).

e. Recommendation [1 Mark]:
Comparing the 30-day Rupee costs (and annualized rates): Plan A = **Rs 7,250 (8.70% / 11.60%)**, Plan B = **Rs 9,166.67 (11.00%)**, Plan C = **Rs 8,283.33 (9.94% / 14.20%)**, and Plan D = **Rs 17,000 (20.40%)**.
South-West Agro Company should choose **Plan A (Local Bank Trust Receipt Loan)** (or **Plan B** if comparing strictly on average balance rate 11.00% vs. 11.60%), as Plan A has the **lowest total Rupee outlay of Rs 7,250** for the 30-day period!

EXAM HINT TO REMEMBER: *In Q19, compute BOTH the 30-day Rupee cost and the Annualized Percentage Rate (APR) for each plan: Plan A costs the lowest total Rupees (Rs 7,250), while Plan B has the lowest rate on average balance (11.00% vs. 11.60% on Plan A's Rs 750,000 average balance). Mentioning both guarantees 10/10 marks!*

Q20. The common stock of the Draybar Corporation earns Rs 2.50 per share, has a dividend payout of two-thirds, and sells at a P/E ratio of 16. Draybar wishes to offer Rs 10 million of 9 percent, 20-year convertible debentures with an initial conversion premium of 20 percent and a call price of 105. Draybar currently has 1 million common shares outstanding and has a 50 percent tax rate. a. What is the conversion price? b. What is the conversion ratio per Rs 1,000 debentures? c. What is the initial conversion value of each debentures? d. How many new shares of common must be issued if all debentures are converted? e. If Draybar can increase operating earnings by Rs 1 million per year with the proceeds of the debenture issue, compute the new earnings per share and earnings retained before and after conversion.

Answer (Step-by-Step Numerical Solution):
Given:
• Current EPS = Rs 2.50; Dividend payout ratio = 2/3 (so current DPS = 2/3 x Rs 2.50 = **Rs 1.6667 per share**, and retention ratio = 1/3)
• Price-Earnings (P/E) ratio = 16x => **Current Market Price of Stock (P_0) = EPS x P/E = Rs 2.50 x 16 = Rs 40.00 per share**
• Convertible Debenture Issue = Rs 10,000,000 (10,000 debentures of Rs 1,000 par value); Coupon = 9%; Initial conversion premium = 20%
• Existing common shares (N_0) = 1,000,000 shares; Corporate tax rate (T) = 50%
• Current Net Income = 1,000,000 x Rs 2.50 = Rs 2,500,000 => **Current Operating Earnings (EBIT, with 0 prior debt and 50% tax) = Rs 2,500,000 / (1 - 0.50) = Rs 5,000,000**.

a. Conversion Price (P_c):
P_c = Current Stock Price (P_0) x (1 + Conversion Premium)
P_c = Rs 40.00 x (1 + 0.20) = **Rs 48.00 per share**

b. Conversion Ratio per Rs 1,000 Debenture (CR):
CR = Face Value / Conversion Price = Rs 1,000 / Rs 48 = **20.8333 shares per debenture**

c. Initial Conversion Value of Each Debenture (CV_0):
CV_0 = Conversion Ratio x Current Stock Price = 20.83333 x Rs 40 = **Rs 833.33 per debenture**

d. Number of New Common Shares Issued If All Debentures Are Converted:
New Shares = Total Debenture Issue / Conversion Price = Rs 10,000,000 / Rs 48 = **208,333.33 shares (or 208,333 shares)**

e. New EPS and Earnings Retained BEFORE and AFTER Conversion (When EBIT Increases by Rs 1,000,000 to Rs 6,000,000):

Income & Retention Particulars	Before Conversion (Rs)	After Full Conversion (Rs)
Old Operating Earnings (EBIT)	5,000,000	5,000,000
Add: Increase in Operating Earnings	1,000,000	1,000,000
New Operating Earnings (EBIT)	**6,000,000**	**6,000,000**
Less: Debenture Interest (9% of Rs 10,000,000)	900,000	0 (Converted)
Earnings Before Taxes (EBT)	5,100,000	6,000,000
Less: Corporate Income Tax @ 50%	2,550,000	3,000,000
Net Income Available to Common (EAT)	**2,550,000**	**3,000,000**
Number of Common Shares Outstanding	1,000,000	1,208,333.33
New Earnings Per Share (EPS)	**Rs 2.55**	**Rs 2.48**
Less: Dividends (Method 1: Constant DPS = Rs 1.6667/sh)	1,666,667	2,013,889
Earnings Retained (Method 1: Constant DPS = Rs 1.6667)	**Rs 883,333**	**Rs 986,111**
Earnings Retained (Method 2: Constant 1/3 Retention Ratio)	**Rs 850,000**	**Rs 1,000,000**

EXAM HINT TO REMEMBER: *In Q20 (Draybar Corp — classic Van Horne problem): P_0 = EPS x P/E = 2.50 x 16 = Rs 40; P_c = 40 x 1.20 = Rs 48. Current NI = Rs 2.5M => Current EBIT (at 50% tax) = Rs 5.0M, so New EBIT = Rs 6.0M! Show both Retention methods (constant DPS Rs 1.6667 -> Retained Rs 883,333 & Rs 986,111; and constant 1/3 retention -> Rs 850,000 & Rs 1,000,000)!*

Q21. What is financial distress? Briefly explain its impact on a firm's operation.

Answer:

1. Meaning of Financial Distress:
Financial distress is a condition in which a corporation experiences severe cash-flow shortages that make it unable (or barely able) to meet its promised obligations to creditors (interest, principal, supplier payables, leases, and wages) as they fall due. If not resolved through operational turnaround or private debt restructuring, financial distress leads to technical default, insolvency, and legal bankruptcy (reorganization or liquidation).

2. Impact of Financial Distress on a Firm's Operations (Direct & Indirect Costs):

• 1. Loss of Customers and Declining Sales (Customer Defection):
Customers become reluctant to buy durable goods, machinery, software, or services from a financially distressed firm because they fear the company will not stay in business to honor warranties, provide spare parts, or deliver future upgrades.

• 2. Tightening of Trade Credit by Suppliers:
Suppliers who fear non-payment stop extending normal trade credit terms (e.g., net 30 or net 60) and demand Cash on Delivery (COD) or advance payment, worsening the firm's liquidity crunch and disrupting raw material supply.

• 3. Loss of Key Employees and Management Distraction:
Talented employees and managers leave for more stable competitors. Senior executives spend most of their time negotiating with banks and creditors rather than managing production, innovation, and marketing.

• 4. Suboptimal Investment Decisions (Underinvestment & Asset Fire Sales):
A distressed firm is forced to pass up profitable positive-NPV projects, slash R&D, defer critical maintenance, and sell valuable core assets at steep "fire-sale" discounts to raise emergency cash.

• 5. Agency Conflicts (Risk Shifting & Milking the Property):
Shareholders of a near-bankrupt firm may be tempted to gamble on overly risky projects ("betting the company" with creditors' money) or pay out liquid assets.

• 6. Direct Legal and Administrative Costs:

High legal fees, accounting/advisory fees, and court expenses if distress escalates into formal insolvency proceedings.

Group D: Comprehensive Answer Questions [20 Marks]

Q22. Case Study: Fewa Securities Inc. (Pokhara Home Office Market Data & Quotation System) Equipment cost = Rs 500,000; 4-year term loan at 10% interest rate; Straight-line depreciation over 4-year life; Expected salvage value = Rs 50,000. Alternative: 4-year lease with payments of Rs 200,000 at the end of each year. Fewa's tax rate = 40%. a. Who are the two parties to a lease contract? b. Describe the various types of leases. c. What is Fewa's present value of cost of leasing the equipment? What is the present value of cost of borrow and buy equipment? Would you recommend to purchase or lease the asset? d. How do you treat repair and maintenance costs in lease versus purchase decision?

Answer (Comprehensive Case Study Solution):

a. The Two Parties to a Lease Contract:

• **1. Lessor (Owner):** The party (equipment manufacturer, leasing company, or bank) that legally owns the asset and grants the right to use it to another party in exchange for periodic lease rental payments (here, Consolidated Leasing / the equipment manufacturer).

• **2. Lessee (User):** The party that acquires the right to possess and use the leased asset for a specified period by making periodic lease payments to the lessor (here, **Fewa Securities Inc.**).

b. Various Types of Leases:

• **1. Operating Lease:** A short-term, cancellable lease for a period shorter than the asset's economic life, where the lessor bears maintenance, insurance, and obsolescence risk.

• **2. Financial (Capital) Lease:** A long-term, non-cancellable lease that fully amortizes the lessor's cost of the asset and transfers maintenance, insurance, and ownership risks to the lessee.

• **3. Sale and Leaseback Arrangement:** A firm sells an asset it already owns (such as a building or computer system) to a lessor for immediate cash and simultaneously leases it back for long-term use.

• **4. Leveraged Lease:** A three-party lease involving a lessee, an equity lessor (who puts up 20%–40% of the cost), and a third-party lender (who provides 60%–80% non-recourse debt).

• **5. Direct Lease:** A lease in which the lessee acquires the use of a new asset it did not previously own directly from the manufacturer or leasing company.

c. Present Value of Cost of Leasing vs. Cost of Borrow-and-Buy and Recommendation:

Given:

• Cost of equipment (I_0) = Rs 500,000; Useful life (n) = 4 years; Salvage value (SV) = Rs 50,000

• Annual end-of-year lease payment (L) = Rs 200,000; Corporate tax rate (T) = 40%

• Before-tax cost of debt (k_d) = 10% => **After-tax cost of debt (k_dt) = 10% x (1 - 0.40) = 6%**

• Present Value Factors at 6% for 4 years:

- PVIFA(6%, 4 yrs) = [1 - (1.06)^(-4)] / 0.06 = **3.465106**

- PVIF(6%, 4 yrs) = (1.06)^(-4) = **0.792094**

• **1. Present Value of Cost of Leasing:**

- Annual after-tax lease payment = Rs 200,000 x (1 - 0.40) = **Rs 120,000 per year**

- **PV of Cost of Leasing** = Rs 120,000 x PVIFA(6%, 4 yrs) = Rs 120,000 x 3.465106 = **Rs 415,812.67** (or Rs 415,812 using 3.4651).

• **2. Present Value of Cost of Borrowing and Buying (Purchasing):**

- Annual Straight-Line Depreciation = (Rs 500,000 - Rs 50,000) / 4 = **Rs 112,500 per year**

- Annual Depreciation Tax Shield = Rs 112,500 x 0.40 = **Rs 45,000 per year**

- Terminal Salvage Value in Year 4 = Rs 50,000 (equals Book Value, so no salvage tax)

- **PV of Cost of Purchasing** = Initial Cost - PV(Depreciation Tax Shield) - PV(Salvage Value)

= Rs 500,000 - (Rs 45,000 x 3.465106) - (Rs 50,000 x 0.792094)

= Rs 500,000 - Rs 155,929.75 - Rs 39,604.68 = **Rs 304,465.56** (or Rs 304,466 using 4-decimal tables).

Year	Loan PMT (500k / 3.1699)	Interest @ 10% (Rs)	Depreciation (Rs)	Tax Shield 40% x (Int+Dep)	Salvage Inflow (Rs)	Net Cash Outflow (Rs)	PV @ 6% (Rs)
1	157,735.40	50,000.00	112,500.00	65,000.00	0	92,735.40	87,486.23
2	157,735.40	39,226.46	112,500.00	60,690.58	0	97,044.82	86,369.54
3	157,735.40	27,375.57	112,500.00	55,950.23	0	101,785.17	85,460.75

Year	Loan PMT (500k / 3.1699)	Interest @ 10% (Rs)	Depreciation (Rs)	Tax Shield 40% x (Int+Dep)	Salvage Inflow (Rs)	Net Cash Outflow (Rs)	PV @ 6% (Rs)
4	157,735.40	14,339.58	112,500.00	50,735.83	50,000.00	56,999.57	45,149.04
Total PV of Cost of Purchasing							**304,465.56**

• **3. Recommendation:**

Since the Present Value of Cost of Borrowing and Buying (**Rs 304,465.56**) is substantially lower than the Present Value of Cost of Leasing (**Rs 415,812.67**) by **Rs 111,347.11**, Fewa Securities Inc. should **BORROW AND PURCHASE the equipment**.

d. Treatment of Repair and Maintenance Costs in Lease vs. Purchase Decisions:

• Case 1 (When Lessee Pays Maintenance Under Both Lease and Purchase — Net/Financial Lease):

If the lessee is responsible for paying repair and maintenance costs under both alternatives, the maintenance cash flows are identical and **relevant/incremental cost is zero (ignored in the differential comparison)**.

• Case 2 (When Lessor Pays Maintenance Under the Lease, While Buyer Must Pay Maintenance If Purchased — Full-Service/Operating Lease):

Since repair and maintenance expenses are tax-deductible operating expenses, if the firm buys the asset it must pay an extra **After-Tax Maintenance Cost = Maintenance Expense x (1 - T)** each year. Therefore, in the analysis:

- Either **add the Present Value of After-Tax Maintenance Outflows [Maintenance x (1 - T) x PVIFA(k_dt, n)] to the Cost of Purchasing**,
- Or equivalently, **subtract the after-tax maintenance savings from the Cost of Leasing**.

EXAM HINT TO REMEMBER: *In Q22(d), remember the tax shield on maintenance: if the Lessor pays maintenance under the lease, the Buyer faces an incremental after-tax outflow of Maintenance x (1 - T) each year, which is discounted at k_d(1 - T) and ADDED to the Cost of Purchasing!*

PAPER 3: SHANKER DEV CAMPUS (TU) — BNK 209: INSURANCE AND RISK MANAGEMENT

BBA / Eighth Semester | Preboard Examination 2026 | Full Marks: 100 | Time: 3 Hrs

Group A: Brief Answer Questions [10 x 2 = 20]

Q1. Differentiate between risk and uncertainty.

Answer:

• **Risk:** A situation where future outcomes are unknown, but the underlying **probabilities of possible outcomes can be objectively measured or estimated** using historical data (measurable, objective, and insurable).

• **Uncertainty:** A subjective state of mind characterized by doubt about the future where **probabilities of outcomes cannot be objectively measured or quantified** due to lack of historical data (unmeasurable and generally uninsurable).

Q2. What is a hazard in insurance?

Answer: In insurance, a **hazard** is any condition or situation that creates or increases the probability (frequency) or severity of a loss arising from a given peril. There are four major types of hazards: **(1) Physical Hazard** (icy roads, faulty wiring), **(2) Moral Hazard** (dishonesty/fraud), **(3) Attitudinal / Morale Hazard** (carelessness because of insurance), and **(4) Legal Hazard**.

Q3. Distinguish between named-perils coverage and open-perils coverage.

Answer:

• **Named-Perils (Specified-Perils) Coverage:** An insurance policy that covers losses caused **only by the specific perils explicitly listed (named) in the policy** (such as fire, lightning, windstorm, or hail); the burden of proof is on the insured to show a named peril caused the loss.

• **Open-Perils (All-Risks / Special) Coverage:** An insurance policy that covers losses from **all accidental causes of loss EXCEPT those specifically excluded** in the policy; the burden of proof is on the insurer to prove an exclusion applies.

Q4. Explain the concept of pooling of losses.

Answer: Pooling of losses (risk pooling) is the heart of insurance: it is the spreading of unexpected financial losses incurred by a few unfortunate members over the entire group of insured exposure units so that, in the process, **average loss is substituted for actual loss**. By combining a large number of independent units under the Law of Large Numbers, insurers can predict aggregate losses accurately.

Q5. What is commercial liability insurance?

Answer: Commercial liability insurance (such as a Commercial General Liability or CGL policy) is business insurance that protects an enterprise against legal liability to pay damages and legal defense costs for bodily injury, property damage, personal injury, or defective products/services caused to third parties (customers, visitors, or the public) arising out of business operations.

Q6. Explain the meaning of 'fire' in the context of fire insurance.

Answer: For a loss to be covered under a standard Fire Insurance Policy, "fire" must satisfy two strict legal conditions:

• **1. Actual Ignition (Combustion with Flame/Glow):** There must be actual ignition or burning—mere heating, scorching, charring, or fermentation without flame is not "fire."

• **2. Accidental / Fortuitous ("Hostile Fire" vs. "Friendly Fire"):** The fire must be accidental (not deliberate arson by the insured) and must be a **hostile fire** that escapes its intended container (unlike a *friendly fire* burning safely inside a stove or furnace).

Q7. Explain risk avoidance with an example.

Answer: Risk avoidance is a proactive risk-control technique in which an individual or business completely refuses to engage in an activity, or abandons an existing activity/asset, so that the chance of loss from that exposure is reduced to **zero (0% frequency)**.

• **Example:** A pharmaceutical company deciding not to manufacture a drug with severe side-effect lawsuit risks, or a firm choosing not to build a warehouse in an active flood plain.

Q8. How is an endorsement different from a rider?

Answer: Both are written legal amendments attached to an original insurance policy to modify coverage (add benefits, change limits, or add exclusions), but they differ by insurance branch:

• **Endorsement:** The term used in **Property and Casualty (Non-Life / General) Insurance** (e.g., adding an Earthquake

Endorsement to a Fire Policy).
• **Rider:** The term used in **Life and Health Insurance** (e.g., attaching a Critical Illness Rider or Waiver of Premium Rider to a Life Policy).

Q9. Distinguish between risk appetite and risk tolerance.

Answer:
• **Risk Appetite:** The broad, strategic level and types of risk an organization is **willing to pursue or accept** in order to achieve its long-term strategic objectives and returns.
• **Risk Tolerance:** The specific, measurable **maximum boundary or acceptable variation (quantitative deviation)** around specific objectives that the organization can bear without jeopardizing its financial stability.

Q10. What is Value at Risk (VaR) analysis?

Answer: Value at Risk (VaR) is a statistical financial risk management metric that quantifies the **maximum expected financial loss** on a portfolio or enterprise over a specified time horizon (e.g., 1 day, 10 days, or 1 year) at a given confidence level (such as 95% or 99%) under normal market conditions.

Group B: Short Answer Questions (Attempt any SIX — ALL Solved) [6 x 5 = 30]

Q11. Differentiate between pure risk and speculative risk with examples.

Answer:

Basis of Difference	Pure Risk	Speculative Risk
1. Possible Outcomes	Only two possible outcomes: **Loss or No Loss** (zero chance of financial gain).	Three possible outcomes: **Loss, No Loss (Breakeven), or Gain/Profit**.
2. Insurability	Generally **insurable** through commercial insurance contracts.	Generally **not insurable** through traditional insurance; managed via hedging or diversification.
3. Voluntariness	Involuntary and accidental; people do not willingly seek pure risk.	Undertaken voluntarily and deliberately in pursuit of profit.
4. Applicability of Law of Large Numbers	Easily predictable in aggregate using the Law of Large Numbers.	Harder to predict via static actuarial tables due to market cycles.
5. Impact on Society	Always harmful to society when a loss occurs (destroys wealth).	Can benefit society through innovation, price discovery, and economic growth even if some firms lose.
6. Examples	Fire, earthquake, flood, premature death, motor accident, theft.	Investing in shares, launching a new startup, foreign exchange trading, real estate speculation.

Q12. Explain the concept and key features of commercial property insurance.

Answer:
Concept of Commercial Property Insurance:
Commercial property insurance is a category of non-life business insurance that indemnifies organizations (manufacturers, retailers, banks, service firms) against direct physical loss or damage to their commercial buildings, plant and machinery, office equipment, furniture, and inventory/stock, as well as indirect consequential losses (business interruption) caused by insured perils such as fire, lightning, explosion, natural disasters, and burglary.

Key Features of Commercial Property Insurance:
• **1. Coverage of Direct and Indirect (Consequential) Losses:** Covers both physical repair/replacement of damaged buildings, machinery, and inventory (Direct Loss) and lost net profits plus continuing fixed expenses during the restoration period (Business Interruption / Loss of Profits Insurance).
• **2. Named-Perils vs. Open-Perils (All-Risk) Options:** Can be written on a Standard Fire and Allied Perils basis or on an Industrial All Risks (IAR) open-perils basis.
• **3. Strict Contract of Indemnity & Coinsurance (Average Clause):** Compensation never exceeds the actual financial loss. Under the *Condition of Average (Coinsurance Clause)*, if a business under-insures its property below its true value, claim payouts are reduced proportionately: Claim = Loss x (Sum Insured / True Value).
• **4. Valuation Basis (Actual Cash Value vs. Replacement Cost):** Claims are settled either on Actual Cash Value (Replacement Cost minus Depreciation) or on a Reinstatement / Replacement Cost basis.
• **5. Customizable Endorsements:** Can be tailored with riders/endorsements for Earthquake, Flood, Riot/Strike/Malicious Damage (RSMD), Boiler & Machinery Breakdown, and Electronic Equipment Insurance (EEI).

Q13. Compare and contrast traditional risk management and Enterprise Risk Management (ERM).

Answer:

Dimension	Traditional Risk Management (TRM)	Enterprise Risk Management (ERM)
1. Scope of Risks Covered	Focuses narrowly on **insurable pure risks** (property, liability, employee injury/death).	Covers **all organizational risks**: Pure, Financial/Speculative, Operational, Strategic, Cyber, and Reputational.
2. Organizational Approach	**Fragmented ('Silo') Approach:** Each department manages its own risks independently without coordination.	**Holistic & Integrated Approach:** Manages all risks across the entire enterprise as an interconnected portfolio.
3. View of Risk	Views risk strictly as a **negative threat** to be avoided or insured away.	Views risk as both a **threat to be controlled and a strategic opportunity** to create competitive advantage.
4. Leadership & Governance	Handled by a mid-level Insurance / Risk Manager reporting to the Treasurer.	Led from the top by the **Board of Directors and a Chief Risk Officer (CRO)** aligned with corporate strategy.
5. Primary Risk Tools	Commercial insurance policies, safety inspections, and basic loss retention.	Insurance plus financial derivatives, natural hedges, VaR modeling, stress testing, and strategic diversification.

Q14. State the principle of indemnity and explain its primary purpose in insurance contracts.

Answer:

1. Statement of the Principle of Indemnity:

The Principle of Indemnity states that an insurance contract is a contract of compensation only: in the event of a covered loss, the insurer agrees to pay **no more than the actual financial loss suffered** by the insured, restoring the insured to the approximate financial position they occupied immediately prior to the occurrence of the loss—neither better off nor worse off.

2. Primary Purposes of the Principle of Indemnity:

• **(a) To Prevent the Insured from Profiting from a Loss:** If an insured could collect Rs 5,000,000 on a warehouse worth only Rs 2,000,000, insurance would turn into a lucrative profit-making scheme rather than a loss-restoration mechanism.

• **(b) To Reduce Moral Hazard (Arson and Intentional Fraud):** If policyholders could collect more than the true value of damaged property, dishonest individuals would have a strong financial incentive to intentionally destroy their own property (arson, staged theft, or sinking ships) to pocket the excess insurance money.

• **(c) To Maintain Fair Premium Rates and Social Equity:** By limiting claim payments strictly to actual cash value (Replacement Cost less Depreciation) and enforcing the corollaries of *Subrogation* and *Contribution*, indemnity keeps aggregate claims down and insurance premiums affordable for the public.

Q15. Describe the legal doctrines that support the principle of utmost good faith in insurance contracts.

Answer:

The **Principle of Utmost Good Faith (Uberrimae Fidei)** requires both the insurance applicant and the insurer to observe a high standard of honesty and disclose all material facts truthfully. It is legally enforced through three fundamental legal doctrines:

• **1. Doctrine of Representations:**

- *Meaning:* Representations are oral or written statements made by the applicant on the insurance proposal form before the policy is issued (such as age, health history, smoking habits, or building construction) to give the underwriter information to evaluate the risk.
- *Legal Effect:* If an applicant makes a **material misrepresentation** (a false statement regarding a fact that would have caused the insurer to reject the risk or charge a higher premium), the insurer has the legal right to **void (rescind) the insurance contract** and deny the claim, even if the misrepresentation was unintentional.

• **2. Doctrine of Concealment:**

- *Meaning:* Concealment is the intentional silence or failure of the insurance applicant to disclose a known **material fact** to the insurer when applying for insurance (because the insurer cannot know every hidden risk of the applicant).
- *Legal Effect:* If the insurer proves that the concealed fact was material and that the insured intentionally hid it to obtain insurance (fraudulent concealment), the contract is voidable at the insurer's option.

• **3. Doctrine of Warranties:**

- *Meaning:* A warranty is a formal clause or promise written directly into the insurance policy contract whereby the insured guarantees that a certain condition exists (*affirmative warranty*) or will continuously be maintained during the policy term (*promissory warranty*—e.g., maintaining a working fire sprinkler system or 24-hour armed guard in a bank).

- *Legal Effect:* Unlike representations (which only require substantial truth), a warranty is a strict condition of the contract; any breach of warranty allows the insurer to deny liability.

Q16. Explain the risk control and risk financing techniques that risk managers use to treat loss exposures.

Answer:

Once loss exposures are identified and evaluated, risk managers select from two broad categories of risk treatment techniques:

1. Risk Control Techniques (Aimed at Reducing the Frequency and/or Severity of Losses):

• **(a) Avoidance:** Completely refusing to undertake a risky activity or discontinuing an existing exposure so loss frequency becomes zero (e.g., exiting a hazardous product line).

• **(b) Loss Prevention:** Engineering and behavioral measures designed to reduce the **probability (frequency)** of a loss before it happens (e.g., driver safety training, strict quality control, employee health checkups, firewall installation, no-smoking rules).

• **(c) Loss Reduction (Mitigation):** Measures designed to reduce the **financial severity** of a loss after a peril occurs, including:

- *Pre-loss & Post-loss physical measures:* Automatic fire sprinklers, airbags, emergency evacuation drills, and salvage operations.
- *Duplication:* Keeping spare backup servers or off-site cloud backups.
- *Separation / Diversification:* Spreading inventory across multiple warehouses in different cities so a single fire cannot destroy all stock.

2. Risk Financing Techniques (Arrangements to Pay for Losses That Do Occur):

• **(a) Retention (Active or Passive Self-Insurance):** The organization retains part or all of the financial consequences of a loss exposure (ideal for High-Frequency, Low-Severity predictable losses) funded through current net income, deductibles, earmarked contingency reserves, or a **Captive Insurance Company**.

• **(b) Non-Insurance Risk Transfer:** Shifting the financial burden of loss to a third party other than an insurance company via hold-harmless / indemnity clauses in contracts, leasing agreements, subcontractors, or financial derivatives (hedging).

• **(c) Commercial Insurance:** Transferring Low-Frequency, High-Severity pure risks to a professional insurance company in exchange for a known premium.

Q17. Explain why catastrophic losses pose challenge to the insurance mechanism.

Answer:

Ideally Insurable Risks require that exposure units be **independent (non-correlated)** and that the maximum possible loss not be catastrophic at the same time. **Catastrophic losses**—such as major earthquakes (like the 2015 Gorkha earthquake in Nepal), widespread floods, hurricanes, pandemics, wars, or systemic cyberattacks—pose a severe challenge to the private insurance mechanism for the following reasons:

• **1. Breakdown of the Law of Large Numbers (High Positive Correlation of Exposures):**

Risk pooling works only when losses are independent so that only a tiny fraction of policyholders claim in any year. In a catastrophe (such as an earthquake or flood across Kathmandu Valley), thousands of insured homes, factories, and vehicles are damaged **simultaneously** by a single event, invalidating the independence assumption.

• **2. Threat of Insurer Insolvency and Liquidity Crisis:**

When hundreds of thousands of huge claims arrive at once, total claim payouts can exceed the insurer's annual premium income and capital reserves, forcing insurers to liquidate long-term investments at fire-sale prices or go bankrupt.

• **3. Extreme Difficulty in Actuarial Rate Making (Unpredictable Frequency & Severity):**

Because mega-catastrophes occur rarely and irregularly (e.g., once every 50–100 years) and property values/climate patterns shift rapidly, historical data is insufficient to calculate a stable annual pure premium.

• **4. Affordability vs. Availability Dilemma (Adverse Selection):**

To cover catastrophic tail risk, insurers must charge high catastrophe loadings; then only people living in extreme danger zones (e.g., floodplains or fault lines) buy coverage, worsening adverse selection.

• **5. Need for Global Reinsurance, Catastrophe Bonds, and Government Backstops:**

Domestic insurers cannot absorb catastrophic losses alone and must rely heavily on international reinsurance treaties, catastrophe (CAT) pools, and government disaster funds.

Group C: Long Answer Questions (Attempt any THREE — ALL Solved) [3 x 10 = 30]

Q18. Explain the concept of loss exposure and its various types with examples.

Answer:

1. Concept of Loss Exposure:

In risk management, a **loss exposure** is defined as any situation, asset, activity, or circumstance in which a possible financial loss exists—regardless of whether an actual loss occurs or not. Every loss exposure consists of three elements: (i) an *Asset or Subject Matter at value*, (ii) a *Peril* that can cause loss, and (iii) the *Financial Consequences* if the loss happens.

2. Various Types of Loss Exposures with Examples:

• 1. Property Loss Exposures (Direct and Indirect):

- *Direct Property Loss Exposure:* The risk of physical damage to, destruction of, or theft of tangible or intangible property owned by a firm or family.

Examples: A factory building damaged by fire or earthquake; delivery trucks damaged in a collision; raw material inventory ruined by warehouse flooding; office laptops stolen in a burglary.

- *Indirect (Consequential / Business Interruption) Loss Exposure:* Financial losses resulting from the inability to use damaged property after a direct physical loss.

Examples: Loss of net profit while a burned-down hotel is being rebuilt; extra expense of renting temporary office space and generators after a flood.

• 2. Liability Loss Exposures (Third-Party Legal Liability):

The risk of being held legally liable by a court of law to pay compensatory/punitive damages and legal defense costs for bodily injury, property damage, or financial harm caused to third parties.

Examples:

- *Premises and Operations Liability:* A customer slipping and breaking a leg on a wet supermarket floor.
- *Products Liability:* A food manufacturer sued because contaminated food sickened consumers.
- *Motor / Auto Third-Party Liability:* A company delivery van hitting a pedestrian.
- *Professional / Errors & Omissions (E&O) Liability:* A surgeon sued for malpractice, or an IT firm sued for a client data breach.
- *Employers' Liability / Workers' Compensation:* Factory workers injured by machinery on the job.

• 3. Human Resource (Personnel) Loss Exposures:

Financial losses suffered by a family or business due to the death, disability, illness, retirement, or resignation of key individuals or employees.

Examples: Premature death of a family breadwinner; sudden death or disability of a startup's chief software architect (Key-Person Risk); workplace accidents or serious illnesses of employees.

• 4. Crime, Fidelity, and Cyber Loss Exposures:

Losses resulting from dishonest or criminal acts by employees or external criminals.

Examples: Employee embezzlement of cash (fidelity exposure), robbery of bank cash-in-transit, ransomware attacks, and theft of intellectual property.

• 5. Financial and Failure-to-Perform (Credit) Loss Exposures:

Losses caused by external market fluctuations or default by business partners.

Examples: Default by a major credit customer on accounts receivable, contractor failure to finish a project on time, or adverse interest/exchange rate swings.

Q19. Discuss the factors influencing rate fixation in fire insurance and explain the methods used to determine rates.

Answer:

Part 1: Major Factors Influencing Rate Fixation (Underwriting & Rating Factors) in Fire Insurance:

When fixing the premium rate for a building and its contents under fire insurance, insurers and tariff boards evaluate physical and moral hazards across the following key factors:

• 1. Construction Material and Structural Design (Class of Construction):

Reinforced Cement Concrete (RCC / Class A fire-resistive) buildings have much lower fire spread risk and receive lower premium rates than brick-and-timber or thatched/wooden structures. Building height, number of stories, and fire-proof partition walls also affect the rate.

• 2. Occupancy (Nature of Business / Use of Building):

What happens inside the building is critical: a residential house, school, or bank office carries very low fire hazard compared to a garment factory, chemical/paint plant, sawmill, or fireworks warehouse storing flammable materials.

• 3. Location and Exposure Hazard (Surrounding Neighborhood):

Whether the building is located in a congested market area next to hazardous factories (high exposure hazard from neighboring fires) or in an open area with wide road access for fire engines.

• 4. Internal Fire Protection and Loss-Prevention Appliances:

Buildings equipped with automatic fire sprinklers, smoke detectors, fire alarms, hydrants, fire extinguishers, and trained fire-safety staff receive substantial tariff discounts.

• **5. Public Fire-Fighting Facilities:**
Proximity to a municipal fire brigade station and availability of reliable water mains/hydrants.

• **6. Past Loss Experience and Moral Hazard of the Insured:**
Housekeeping standards, electrical wiring maintenance, and the insured's past claims record and financial integrity.

Part 2: Methods Used to Determine Rates in Fire Insurance:

• **1. Judgment Rating (Individual / Specific Evaluation):**
Each property exposure is evaluated individually by an experienced underwriter based largely on professional judgment without a formal mathematical table. Used when risks are highly unique or non-standardized (e.g., specialized historic monuments or complex offshore plants).

• **2. Class Rating (Manual / Tariff Rating):**
Exposures with similar characteristics (such as residential dwellings, retail shops, or standard offices of the same construction class and location) are grouped into homogeneous rating classes, and every unit in that class is charged the same standardized manual rate per Rs 1,000 of sum insured. It is simple, fast, and inexpensive to administer.

• **3. Schedule Rating (Physical Hazard / Analytic Schedule Rating):**
Widely used for commercial and industrial fire insurance: the underwriter starts with a **baseline "basis rate"** for a standard building and then systematically **adds debits (surcharges)** for substandard physical features (e.g., flammable storage, faulty wiring, lack of firewalls) and **subtracts credits (discounts)** for superior fire-protection features (e.g., automatic sprinklers, fire alarms, watchmen). This directly incentivizes business owners to install fire-safety equipment.

• **4. Experience / Merit Rating (for Large Industrial Risks):**
Adjusts the standard class or schedule rate upward or downward based on the insured's own historical fire loss experience over recent years.

Q20. Compare and contrast the term life insurance and cash-value life insurance.

Answer:

Life insurance policies are broadly divided into two major categories: **Term Life Insurance** (pure protection) and **Cash-Value Life Insurance** (permanent/endowment protection bundled with savings). Their similarities and differences are detailed below:

1. Similarities Between Term Life and Cash-Value Life Insurance:

• Both are legal life insurance contracts governed by the principles of insurable interest and utmost good faith.

• Both pay a tax-exempt lump-sum death benefit to the named beneficiary/nominee if the insured dies while the policy is in force.

• Both allow attachment of optional riders (Accidental Death & Dismemberment, Permanent Total Disability, Critical Illness, Waiver of Premium) and qualify for income tax deduction on premiums paid (up to Rs 40,000 p.a. in Nepal).

2. Detailed Comparison (Contrasts) Between Term Life and Cash-Value Life Insurance:

Basis of Comparison	Term Life Insurance	Cash-Value Life Insurance (Whole Life / Endowment)
1. Duration of Coverage	**Temporary:** Covers a specific period (e.g., 1, 5, 10, 20, or 30 years) and expires at the end of the term.	**Long-Term / Permanent:** Covers the entire lifetime (Whole Life) or pays out at a set maturity date (Endowment).
2. Savings / Cash Value Component	**None (Pure Protection Only):** Builds zero cash surrender value; if the insured survives the term, nothing is paid back.	**Includes Savings Element:** Accumulates a guaranteed **Cash Surrender Value** plus vested bonuses over time.
3. Premium Cost per Rs 1,000 Coverage	**Very Low Initial Premium:** Gives maximum death benefit per Rupee of premium for young families.	**Much Higher Level Premium:** Because part of the premium funds the savings/investment reserve.
4. Policy Loans & Surrender	Cannot be surrendered for cash and cannot be pledged for a policy loan.	Policyholder can surrender the policy for cash or borrow up to 80%–90% of the accumulated cash value.
5. Maturity Benefit	Pays **only upon death** during the policy term (no survival benefit unless a Return-of-Premium rider is bought).	Pays upon death OR upon survival to maturity (Endowment) or age 100 (Whole Life).

Basis of Comparison	Term Life Insurance	Cash-Value Life Insurance (Whole Life / Endowment)
6. Ideal Target Customer	Young breadwinners with high debt/family obligations and limited cash flow (or savvy investors who 'Buy Term and Invest the Difference').	Individuals seeking disciplined long-term forced savings, retirement/education corpus, or permanent estate liquidity.

Q21. Discuss loss forecasting as a fundamental activity in risk management.

Answer:

1. Concept and Importance of Loss Forecasting in Risk Management:

Loss forecasting is the quantitative cornerstone of Step 3 of the risk management process (Risk Evaluation and Measurement). After identifying loss exposures, the risk manager must predict the **expected frequency** (how many losses will occur) and **expected severity** (the average and maximum Rupee size of losses) over the upcoming budget period. Loss forecasting is fundamental because a risk manager cannot rationally decide whether to *retain* a risk, *buy commercial insurance*, choose a *deductible level*, or invest in costly *loss-prevention equipment* without first estimating the expected cost and variability of future losses.

2. Major Quantitative Techniques of Loss Forecasting:

• **1. Probability Analysis:**

- Uses historical loss records to estimate the objective probability of single or multiple independent and dependent events.
- By multiplying each possible loss outcome by its probability, the risk manager computes the **Expected Value (Expected Loss = Frequency x Severity)** and the **Standard Deviation** (measuring objective risk/uncertainty around the average loss).

• **2. Regression Analysis (Causal / Trend Forecasting):**

- Establishes a mathematical relationship between a *dependent variable* (such as annual number of workers' compensation claims or fleet collision losses) and one or more *independent physical exposure variables* (such as total payroll hours, number of employees, production volume, or total vehicle kilometers driven): **Y = a + bX**.
- As the firm budgets its planned payroll or vehicle mileage (X) for next year, regression predicts the expected claims (Y).

• **3. Forecasting Based on Theoretical Loss Distributions (Simulation):**

- Fits empirical loss data to theoretical probability distributions: typically, the **Poisson or Binomial distribution** is used to model *loss frequency*, and the **Lognormal or Pareto distribution** is used to model *loss severity*.
- Using Monte Carlo simulation, risk managers estimate the Probable Maximum Loss (PML) and Value at Risk (VaR) at 95% or 99% confidence levels.

3. Role of Loss Forecasting in Risk Management Decision-Making:

• Determines accurate self-insurance reserve budgets and captive capital requirements.
• Evaluates whether an insurer's quoted commercial premium is fair relative to the firm's own expected losses.
• Enables cost-benefit (NPV) analysis of safety and loss-control investments.

Group D: Comprehensive Answer Questions [4 x 5 = 20]

Q22. Case Study: Shree Sharma (35-Year-Old Software Engineer in Kathmandu) Shree Sharma (age 35, married software engineer, sole breadwinner for wife age 32, daughter age 8, and dependent mother age 68; expects to retire at age 60). Annual salary = NPR 18,00,000; after taxes and personal living expenses, NPR 12,00,000 per year is available for family's support. Financial obligations & goals: Outstanding home loan = NPR 50,00,000 (to be paid off immediately upon death); Daughter's higher-education fund (required 10 years from now) = NPR 30,00,000; Emergency fund needed = NPR 10,00,000; Monthly family living expenses = NPR 60,000 (NPR 7,20,000/year); Wife has no independent income now but could potentially earn after 2 years. Annual discount rate = 7%. a. Using the Human Life Value (HLV) Approach, estimate the approximate amount of life insurance Shree should own. Show the logic of your calculation clearly. b. Apply the Needs Approach to determine Shree's total insurance need. You may make reasonable assumptions. c. Based on Shree's age, family situation, and financial goals, recommend and justify the type of life insurance policy you consider most suitable for him (e.g., Term, Whole Life, Endowment, or Universal Life). Explain any relevant riders you would recommend attaching to the policy. d. The insurance company informs Shree that premiums for the suggested policy are calculated using the equivalence principle. The assumed interest rate is 7%, and the mortality table shows a probability of death of 0.0020 for a 35-year-old male. If Shree wishes to purchase a 1-year term policy with a face value of NPR 10,00,000, calculate the Net Single Premium (NSP) for this one-year policy.

Answer (Step-by-Step Comprehensive Case Solution):

a. Estimation of Life Insurance Using the Human Life Value (HLV) Approach [5 Marks]:
Logic & Given Data:
• Current age = 35 years; Planned retirement age = 60 years => **Remaining working years (n) = 60 - 35 = 25 years**
• Annual gross salary = NPR 18,00,000
• Net annual income available for family support (after taxes and Shree's personal living expenses) = **NPR 12,00,000 per year**
• Discount rate (i) = 7% p.a.
• **Present Value Interest Factor of Annuity for 25 Years at 7% [PVIFA(7%, 25 yrs)]:**
PVIFA(7%, 25) = [1 - (1 + 0.07)^(-25)] / 0.07 = (1 - 0.184249) / 0.07 = **11.653583**
• **Human Life Value (HLV):**
HLV = Annual Family Share x PVIFA(7%, 25 yrs)
HLV = NPR 12,00,000 x 11.653583 = **NPR 1,39,84,300 (or approx. NPR 13,984,300 / NPR 1.40 Crore)**
*(Note: If annuity due [beginning-of-year] is used: 13,984,300 x 1.07 = **NPR 1,49,63,201**.)*

b. Estimation of Total Insurance Need Using the Financial Needs Approach [5 Marks]:
Under the **Needs Approach**, total life insurance required equals the Present Value of Lump-Sum Cash Needs plus the Present Value of Ongoing Family Income Needs (minus existing liquid financial assets = NPR 0):
• **1. Immediate Lump-Sum Cash & Goal Needs:**
- (i) **Mortgage / Home Loan Payoff Fund: NPR 50,00,000**
- (ii) **Emergency & Final Clearance Fund: NPR 10,00,000**
- (iii) **Daughter's Higher-Education Fund (NPR 30,00,000 required 10 years from now):**
• *Method A (Discounted Present Value at 7% for 10 years):*
PV = NPR 30,00,000 / (1.07)^10 = 30,00,000 x 0.508349 = **NPR 15,25,048**
• *Method B (Full Unadjusted Face Value Reserved Today, assuming 7% return offsets 7% education inflation):* **NPR 30,00,000**

• **2. Present Value of Ongoing Family Living Expenses (NPR 60,000/month = NPR 7,20,000/year):**
- *Standard Assumption 1 (Providing full NPR 7,20,000/year for 25 years until Shree's normal retirement/mother's & family's long-term support at 7%):*
PV(Income Need, 25 yrs) = NPR 7,20,000 x PVIFA(7%, 25 yrs) = 7,20,000 x 11.653583 = **NPR 83,90,580**
- *Alternative Assumption 2 (Providing full NPR 7,20,000/yr for 10 years until daughter turns 18 [PVIFA(7%,10) = 7.02358 -> NPR 50,56,979], or full NPR 7,20,000/yr for 2 years [PVIFA(7%,2) = 1.8080 -> NPR 13,01,773] followed by 50% supplementation [NPR 3,60,000/yr] for years 3–25 after wife takes up employment).*

• **3. Total Life Insurance Need Summary Table (Showing Both Discounted & Nominal Education Fund):**

Needs Category	Case 1: PV of Education + 25-Yr Family Income (NPR)	Case 2: Nominal Education + 25-Yr Family Income (NPR)	Case 3: PV Education + 10-Yr Dependency Period (NPR)
1. Outstanding Home Loan Payoff	50,00,000	50,00,000	50,00,000
2. Emergency Fund Needed	10,00,000	10,00,000	10,00,000
3. Daughter's Education Fund	15,25,048 (PV @ 7%, 10y)	30,00,000 (Nominal)	15,25,048 (PV @ 7%, 10y)
4. PV of Family Living Expenses (7,20,000/yr)	83,90,580 (25 yrs @ 7%)	83,90,580 (25 yrs @ 7%)	50,56,979 (10 yrs @ 7%)
Total Life Insurance Need	**NPR 1,59,15,628**	**NPR 1,73,90,580**	**NPR 1,25,82,027**

c. Recommended Life Insurance Policy & Riders for Shree Sharma [5 Marks]:
• **Recommended Policy:** A **25-Year Level Term Life Insurance Policy** (for the bulk of the coverage, e.g., NPR 1.20 Crore to 1.50 Crore) optionally paired with a **10-Year Child Endowment Policy of NPR 30,00,000** for his daughter's higher education.
• **Justification:** At age 35, Shree has three dependents (wife, 8-year-old daughter, and 68-year-old mother) and a huge immediate NPR 50,00,000 home loan. Term Life Insurance gives him the **highest lump-sum death benefit at the lowest annual premium**, ensuring the mortgage and living expenses are 100% covered without straining his monthly budget. Meanwhile, a 10-year Child Endowment policy guarantees the NPR 30,00,000 college fund in 10 years whether Shree lives or dies.
• **Recommended Riders to Attach:**
1. *Accidental Death and Permanent Total Disability (AD&D / PTD) Rider:* Pays an additional benefit / regular income if an accident leaves Shree permanently disabled.
2. *Critical Illness (CI) Rider:* Provides an immediate lump sum upon diagnosis of major illnesses (heart attack, cancer, stroke, kidney failure) so the family does not have to dip into the home or education fund for hospital bills.

3. *Waiver of Premium Rider:* Waives all future premiums if disability or critical illness prevents Shree from working.

d. Calculation of Net Single Premium (NSP) for a 1-Year Term Policy of NPR 10,00,000 [5 Marks]:

Given:

• Face value (Sum Assured) of 1-year term policy = NPR 10,00,000
• Probability of death at age 35 (q_35) = 0.0020
• Assumed annual interest rate (i) = 7% = 0.07

Step-by-Step Calculation Using the Actuarial Equivalence Principle:

Under the equivalence principle, the Net Single Premium (NSP) paid at the beginning of the year equals the Present Value of the expected death benefit paid at the end of the year:

• Expected Death Claim at Year-End = Face Value x Probability of Death (q_35)
= NPR 10,00,000 x 0.0020 = **NPR 2,000.00**
• Present Value of Expected Claim (Net Single Premium, NSP):
NSP = Expected Claim / (1 + i)^1 = NPR 2,000 / (1 + 0.07) = 2,000 / 1.07 = **NPR 1,869.16**

EXAM HINT TO REMEMBER: *In Q22(d), Net Single Premium (NSP) for a 1-year Term Policy is super easy: NSP = (Face Value x Probability of Death) / (1 + Interest Rate) = (10,00,000 x 0.0020) / 1.07 = 2,000 / 1.07 = NPR 1,869.16!*

PAPER 4: SHANKER DEV CAMPUS (TU) — MGT 240: STRATEGIC MANAGEMENT

BBA / Eighth Semester | Preboard Examination 2026 | Full Marks: 100 | Time: 3 Hrs

Group A: Brief Answer Questions [10 x 2 = 20]

Q1. Define strategic management with suitable components.

Answer: Strategic management is the art and science of formulating, implementing, and evaluating cross-functional decisions that enable an organization to achieve its long-term objectives and sustain a competitive advantage.
Core Components: (1) Strategic Intent (Vision, Mission, Objectives), (2) Environmental Analysis (External & Internal Scanning), (3) Strategy Formulation & Choice, (4) Strategy Implementation, and (5) Strategic Evaluation & Control.

Q2. What do you mean by mission statement?

Answer: A **mission statement** is an enduring declaration of an organization's fundamental purpose and reason for existence ("Who we are, what we do, and whom we serve"). It defines the firm's current business scope—its products/services, target markets, core technologies, and commitment to stakeholders—and distinguishes it from other similar enterprises.

Q3. What do you mean by scenario planning?

Answer: Scenario planning is a strategic foresight technique in which planners construct multiple detailed, plausible alternative narratives ("scenarios"—such as best-case, worst-case, and base-case futures) of how uncertain external macro-environmental forces (PESTEL) might unfold, allowing the firm to test strategies and build contingency plans.

Q4. Write about company values with an example.

Answer: Company values (core values) are the deeply ingrained guiding principles, ethical beliefs, and behavioral norms that govern how employees and leaders act and make decisions within an organization.
• **Example:** Apple Inc.'s core values of *innovation, user-centric design perfectionism, privacy, and environmental sustainability*, or a bank's core values of *integrity, customer trust, and transparency*.

Q5. Define hyper-competition.

Answer: Coined by Richard D'Aveni (1994), **hyper-competition** is an environment of intense, rapid, and aggressive competitive maneuvering (across price/quality, timing/know-how, stronghold creation, and deep pockets) in which competitors quickly erode each other's market advantages, making any competitive advantage temporary rather than permanently sustainable.

Q6. Write about strategic group with suitable example.

Answer: A **strategic group** is a cluster of rival firms within the same industry that follow similar business models and competitive strategies along key strategic dimensions (such as price range, product quality, geographic scope, and distribution channels).
• **Example:** In the automobile industry, luxury automakers (Mercedes-Benz, BMW, Audi) form one strategic group, while economy/mass-market carmakers (Suzuki, Hyundai, Tata) form another strategic group.

Q7. What do you mean by unrelated diversification?

Answer: Unrelated (Conglomerate) Diversification is a corporate-level expansion strategy in which a company enters into an entirely new industry or business line that shares **no common markets, products, technologies, or value-chain linkages** with its existing businesses (e.g., an apparel company acquiring a cement factory or hydro-power plant to spread financial risk).

Q8. What is strategic intent?

Answer: Popularized by Gary Hamel and C.K. Prahalad, **strategic intent** is an organization's overarching, ambitious long-term dream and obsession with winning market leadership that stretches the organization's capabilities beyond its current resources. It is expressed through the hierarchy of Vision, Mission, Goals, and Objectives.

Q9. What do you mean by strategic change?

Answer: Strategic change is the proactive or reactive restructuring of an organization's strategy, structure, systems, technology, and corporate culture to realign the firm with shifting external environmental conditions and maintain competitive survival and growth.

Q10. What is threshold resources?

Answer: Threshold resources are the essential minimum tangible and intangible assets (such as basic machinery, working capital, standard IT software, and licensed staff) required for a firm merely to enter an industry, meet minimum customer expectations, and stay in business. They are necessary for survival, but do not confer a competitive advantage.

Group B: Short Answer Questions (Attempt any SIX — ALL Solved) [6 x 5 = 30]

Q11. Define strategic management and discuss the characteristics of Strategic Management.

Answer:

Definition of Strategic Management:

Strategic management is the ongoing process of analyzing the internal and external environments of a firm, formulating corporate, business, and functional strategies, executing those strategies through organizational structure and resource allocation, and evaluating performance to achieve long-term goals and sustainable competitive advantage.

Key Characteristics of Strategic Management:

• **1. Long-Term and Future-Oriented:** Focuses on a 3-to-10+ year horizon, anticipating future environmental trends rather than routine day-to-day operations.
• **2. Holistic and Organization-Wide Scope:** Integrates all functional departments (finance, marketing, operations, HR, R&D) into a unified direction rather than viewing departments in isolation.
• **3. Driven by Top Management & Board of Directors:** Formulated at the apex level with input from line managers.
• **4. Dynamic, Continuous, and Iterative Process:** Not a one-time paper exercise; strategies are continuously monitored and adapted as external markets evolve.
• **5. External-Internal Fit (Open-System Perspective):** Matches the firm's internal Strengths and Weaknesses (SAP) with external Opportunities and Threats (ETOP).
• **6. Major Resource Commitment:** Involves large, long-lasting commitments of capital, technology, and talent.

Q12. What do you mean by vision statement? Discuss the components of vision statement.

Answer:

Meaning of Vision Statement:

A vision statement is an aspirational, future-oriented statement that articulates where the organization wants to be in the long-term future ("What do we want to become?"). It acts as a guiding North Star that inspires employees and stakeholders.

Major Components of an Effective Vision Statement (Collins & Porras Framework):

• **1. Core Ideology (The Enduring Character of the Organization):**
- *(a) Core Values:* The timeless guiding principles and ethical tenets that never change regardless of market fads (e.g., innovation, integrity, customer obsession).
- *(b) Core Purpose:* The deeper fundamental reason for the organization's existence beyond merely making money.
• **2. Envisioned Future (The Audacious Long-Term Goal):**
- *(a) Big, Hairy, Audacious Goal (BHAG):* A bold, clear, and compelling 10-to-30-year goal that stretches the organization (e.g., Microsoft's early vision: "A computer on every desk and in every home").
- *(b) Vivid Description:* A vibrant, engaging mental picture of what it will look and feel like when the organization achieves its BHAG.
• **3. Clarity, Conciseness, and Inspirational Appeal:** Easy to communicate and memorize across all levels of the enterprise.

Q13. What is SAP? Why it is prepared?

Answer:

1. Meaning of Strategic Advantage Profile (SAP):

A **Strategic Advantage Profile (SAP)** is a systematic diagnostic tool and summary table used in internal environmental analysis that evaluates an organization's key functional capabilities (Marketing, Finance, Operations, HR, R&D, Management/MIS) and distinct resources as competitive **Strengths (+)**, **Weaknesses (-)**, or **Neutral (0)** relative to key competitors in the industry.

2. Why SAP is Prepared (Importance & Purpose):

• **(a) Identifying Core Competencies and Competitive Advantages:** Highlights which internal resources and capabilities are superior to rivals (VRIO strengths) so the firm can build its competitive strategy around them.
• **(b) Pinpointing Critical Internal Weaknesses for Correction:** Reveals functional bottlenecks (such as outdated machinery, high debt, or weak distribution) that must be fixed or outsourced.
• **(c) Pairing with ETOP for SWOT / TOWS Strategy Formulation:** When the internal SAP is matched alongside the external *Environmental Threat and Opportunity Profile (ETOP)*, managers can systematically formulate SO, WO, ST, and WT strategies.
• **(d) Guiding Objective Resource Allocation:** Helps top management allocate capital budgets toward high-leverage

strategic capabilities.

Q14. Explain the resource-based view of strategic formulation.

Answer:

Concept of the Resource-Based View (RBV) of Strategy Formulation:

Pioneered by Birger Wernerfelt, C.K. Prahalad, Gary Hamel, and Jay Barney, the **Resource-Based View (RBV)** argues that a firm's long-term competitive advantage and superior profitability are driven primarily by its **unique internal bundle of resources, capabilities, and core competencies**—rather than merely by external industry positioning (the Industrial Organization / I-O view).

Key Elements of RBV in Strategy Formulation:

• **1. Classification of Organizational Resources:**

- *Tangible Resources:* Financial capital, physical plants, land, machinery, and raw materials.
- *Intangible Resources:* Patents, copyrights, brand equity, corporate reputation, organizational culture, and proprietary databases.
- *Organizational Capabilities:* The firm's skill in combining resources through routines and processes to achieve desired outcomes.

• **2. The VRIO Framework for Sustainable Competitive Advantage:**

Under RBV, a resource or capability generates a **sustainable competitive advantage** only if it satisfies all four **VRIO criteria**:

1. **V — Valuable:** Enables the firm to exploit external opportunities, neutralize threats, and create value for customers.
2. **R — Rare:** Controlled by only one or a small number of competing firms.
3. **I — Inimitable (Costly to Imitate):** Competitors cannot easily duplicate it due to unique historical conditions, causal ambiguity, social complexity, or legal patents.
4. **O — Organized to Capture Value:** The firm has the structure, systems, and culture to exploit the resource fully.

• **3. Inside-Out Strategy Formulation Process under RBV:**

Identify internal resources -> Assess capabilities -> Verify VRIO attributes to find Core Competencies -> Formulate strategies that exploit those core competencies in attractive markets -> Upgrade and renew resources continuously (Dynamic Capabilities).

Q15. Define corporate level strategies with suitable examples.

Answer:

Definition of Corporate-Level Strategy:

Corporate-level strategy (grand strategy) is formulated by top management (the Board and CEO) to define the overall scope and direction of the entire multi-business enterprise—answering *"What businesses/industries should we be in, and how should we allocate resources across our business units to maximize overall corporate value?"*

The Four Major Categories of Corporate-Level Strategies with Examples:

• **1. Stability Strategy (Maintaining the Status Quo):**

Continuing current operations in existing products and markets with incremental improvements when the environment is stable or after rapid expansion (includes *No-Change*, *Pause/Proceed-with-Caution*, and *Profit Strategies*).

Example: A regional utility or established cement plant maintaining steady production capacity during a period of economic consolidation.

• **2. Expansion / Growth Strategy (Increasing Scope and Scale):**

- *Concentration (Intensive Growth):* Market Penetration, Market Development, and Product Development (e.g., Wai Wai expanding noodle variants and exporting to new countries).
- *Vertical Integration:* Backward integration (acquiring suppliers) or Forward integration (opening company-owned retail stores, like Apple Stores).
- *Diversification:* Related / Concentric diversification (Apple expanding from Mac computers into iPod, iPhone, iPad, and Apple Watch) or Unrelated / Conglomerate diversification (Chaudhary Group operating in FMCG, banking, hotels, electronics, and telecom).

• **3. Retrenchment / Defensive Strategy (Reducing Scope to Restore Viability):**

Used during financial distress or declining industry demand: *Turnaround* (cost cutting), *Divestment* (selling off an unprofitable SBU), or *Liquidation* (closing down the business).

• **4. Combination Strategy:**

Simultaneously pursuing different grand strategies across different SBUs (e.g., divesting a loss-making division while aggressively expanding a core tech division).

Q16. Discuss leadership capabilities for successful strategy implementation.

Answer:
Successful strategy implementation requires strategic leaders who possess distinct cognitive, interpersonal, and execution capabilities:
• **1. Strategic Visioning and Sense-Making Capability:**
The ability to translate complex strategic plans into a clear, inspiring vision and actionable daily priorities that every employee can understand.
• **2. Change Management and Overcoming Resistance:**
Skill in diagnosing cultural and political resistance to change, communicating transparently, building coalitions, and motivating employees to embrace new processes.
• **3. Emotional Intelligence (EQ) and Interpersonal Communication:**
Self-awareness, empathy, active listening, and conflict-resolution skills needed to align diverse functional heads and teams.
• **4. Resource Mobilization and Decisive Execution:**
The courage to reallocate capital and talent away from legacy projects toward high-priority strategic initiatives and hold managers accountable for results.
• **5. Talent Development and Empowerment Capability:**
Identifying, coaching, and delegating authority to capable lieutenants (such as Steve Jobs recruiting Tim Cook to overhaul Apple's global operations).
• **6. Ethical Integrity and Cultural Stewardship:**
Shaping a high-performance, accountable, and ethical organizational culture by leading by example.

Q17. Differentiate between Vision and Mission statement.

Answer:

Basis of Difference	Vision Statement	Mission Statement
1. Core Question Answered	Answers: **'What do we want to become in the future?'**	Answers: **'What is our business today, and whom do we serve?'**
2. Time Horizon	**Future-oriented** (long-term dream over 5 to 20+ years).	**Present-oriented** leading into the future (current operational scope).
3. Focus & Content	Focuses on aspiration, market leadership position, and desired future state.	Focuses on products/services, target customers, core technology, and stakeholder values.
4. Primary Purpose	To **inspire and stretch** employees and stakeholders toward a shared dream.	To **guide strategic decision-making** and define boundaries of business scope.
5. Stability / Change	Remains relatively timeless until the grand aspiration is achieved.	May be updated periodically as the company diversifies into new products or markets.

Group C: Long Answer Questions (Attempt any THREE — ALL Solved) [3 x 10 = 30]

Q18. What do you mean by Matrix Structure? Discuss the advantages and disadvantages of Matrix Structure.

Answer:
1. Meaning of Matrix Organizational Structure:
A **Matrix Structure** is a hybrid, dual-authority organizational design that combines two structural dimensions simultaneously—typically superimposing horizontal **Project / Product / Geographic Divisions** across vertical **Functional Departments** (such as R&D, Engineering, Marketing, Finance, and HR).
In a matrix organization, employees report to **two bosses simultaneously** (violating Henri Fayol's classical unity-of-command principle):
1. Their **Functional Manager** (who manages technical standards, training, and functional career development), and
2. Their **Project or Product Manager** (who coordinates cross-functional work to deliver a specific project, product, or client solution on time and within budget).

2. Advantages of Matrix Structure:
• **1. Optimal Utilization and Sharing of Scarce Specialists:** Highly skilled specialists (e.g., software architects, R&D scientists, financial analysts) are shared across multiple projects rather than being duplicated in every division.
• **2. Enhanced Cross-Functional Communication and Innovation:** Breaks down vertical functional silos, fostering rapid information sharing and creative problem-solving across engineering, marketing, and operations.
• **3. High Flexibility and Speed of Response:** Project teams can be rapidly assembled, scaled up, or dissolved as market demands, client projects, or technologies change.
• **4. Holistic Strategic Perspective for Middle Managers:** Exposes functional specialists to general management and project commercial realities, developing future top executives.

3. Disadvantages of Matrix Structure:

• **1. Dual Authority Confusion and Role Ambiguity:** Violating the principle of "one employee, one boss" often creates conflicting instructions, priority clashes, and stress for employees caught between functional and project bosses.

• **2. Power Struggles and Turf Wars Between Managers:** Functional managers and project managers may compete over resource allocation, credit, and authority.

• **3. High Administrative Overhead and Slow Decision-Making (Meeting Overload):** Requires extensive coordination meetings, conflict-resolution mechanisms, and dual reporting systems, increasing managerial overhead costs.

• **4. Risk of Groupthink or Shifting Blame:** When a project fails, dual lines of accountability can make it harder to pinpoint responsibility between the functional head and project head.

Q19. Discuss the Porter's five forces model.

Answer:

Developed by Michael E. Porter (1979), the **Five Forces Model** is the premier analytical framework for diagnosing an industry's competitive structure and long-run profit potential. Porter demonstrates that industry attractiveness is shaped by five interacting competitive forces:

• **1. Threat of New Entrants:**

Potential competitors entering an industry bring new capacity and price competition. The severity of this threat depends on the height of **Entry Barriers**:

- Economies of scale and high capital requirements
- Product differentiation, brand loyalty, and customer switching costs
- Proprietary technology, patents, and learning/experience curve advantages
- Access to distribution channels and favorable raw materials
- Government licensing and regulatory policies.

• **2. Bargaining Power of Buyers (Customers):**

Strong buyers squeeze industry margins by demanding lower prices, better quality, and extended credit/service. Buyer power is high when buyers purchase in large volumes, products are undifferentiated commodities, switching costs are low, and buyers can credibly threaten backward integration.

• **3. Bargaining Power of Suppliers:**

Strong suppliers capture industry value by charging higher prices or limiting quality/supply of critical inputs. Supplier power is high when suppliers are concentrated (oligopoly), inputs have no close substitutes, switching costs are high, and suppliers threaten forward integration.

• **4. Threat of Substitute Products or Services:**

Substitutes are products from another industry that satisfy the same customer need (e.g., digital streaming vs. cinema halls, solar power vs. hydro/diesel). Close substitutes cap the maximum price firms in an industry can charge.

• **5. Intensity of Rivalry Among Existing Competitors:**

The degree of head-to-head competition via price discounting, advertising wars, and new product launches. Rivalry is intense when there are many equally balanced competitors, industry growth is slow, fixed/storage costs are high, differentiation is low, and high exit barriers prevent weak firms from leaving.

Q20. Discuss the process of strategic Management with suitable examples.

Answer:

The **Strategic Management Process** is a logical, continuous four-stage cycle through which organizations chart their long-term direction and achieve superior performance:

• **Stage 1: Establishing Strategic Intent & Environmental Analysis (Scanning):**

- *Defining Vision, Mission, and Objectives:* For example, Apple Inc. shifting its mission from personal computers ("Apple Computer, Inc.") to mobile consumer electronics and digital ecosystems ("Apple Inc." in 2007).
- *External Environmental Scanning:* Analyzing macro-factors (PESTEL) and industry forces (Porter's Five Forces) to identify Opportunities and Threats (ETOP)—e.g., a Nepalese commercial bank analyzing NRB monetary policy and digital banking adoption.
- *Internal Organizational Appraisal:* Auditing resources (RBV/VRIO) and value-chain activities to identify Strengths and Weaknesses (SAP).

• **Stage 2: Strategy Formulation and Strategic Choice:**

- Crafting strategic alternatives at three levels:

1. *Corporate Level:* Growth (diversification, vertical integration, mergers), Stability, or Retrenchment.

2. *Business (SBU) Level:* Porter's Generic Strategies—Cost Leadership, Differentiation (e.g., Apple's premium innovation and design differentiation), or Focus.

3. *Functional Level:* Marketing, Operations (e.g., Tim Cook's Just-in-Time supply chain strategy), Finance, R&D, and HR tactics.

- Evaluating options using BCG/GE matrices, Hofer's Matrix, and SWOT, and selecting the best strategy based on Suitability, Feasibility, and Acceptability.

• **Stage 3: Strategy Implementation (Execution):**

- Putting the chosen strategy into action by designing an appropriate organizational structure, allocating capital budgets, establishing functional policies, exercising strategic leadership, and instilling a supportive corporate culture (such as Steve Jobs fostering a culture of perfectionism and innovation at Apple).

• **Stage 4: Strategic Evaluation and Control:**

- Continuously tracking actual results and environmental premises against planned targets (e.g., when Apple's 2008 *MobileMe* launch suffered synchronization failures, Jobs immediately held a Special Alert / Implementation control review, replaced leadership, and transformed it into the successful *iCloud* service).

Q21. Define Hofer's Matrix. Discuss its importance.

Answer:

1. Definition and Concept of Hofer's Matrix (Product-Market Evolution Portfolio Matrix):

Developed by **Charles W. Hofer (1977)** (and later refined by Hofer and Schendel), **Hofer's Matrix** is a **15-cell (5 x 3)** corporate portfolio analysis tool that evaluates a firm's Strategic Business Units (SBUs) using two fundamental dimensions:

• **Vertical Axis — Stage of Product-Market Evolution (Industry Lifecycle Stage, 5 Stages):**

1. *Development Stage*
2. *Growth Stage*
3. *Shakeout Stage*
4. *Maturity / Saturation Stage*
5. *Decline Stage*

• **Horizontal Axis — Competitive Position of the Business Unit (3 Categories):**

1. *Strong Competitive Position*
2. *Average Competitive Position*
3. *Weak Competitive Position*

Inside the 15-cell grid, each SBU is plotted as a circle whose **total size represents the total industry sales volume**, and the **shaded pie wedge inside the circle represents the SBU's market share**.

2. Importance of Hofer's Matrix in Strategic Management:

• **1. Overcomes the Limitations of the Static 2x2 BCG Matrix:**

Whereas the BCG Matrix reduces industry attractiveness to a simple High/Low growth rate, Hofer's Matrix captures all **five distinct lifecycle stages**—especially highlighting the critical **Development** and **Shakeout** stages where strategic mistakes are most common.

• **2. Visualizes Portfolio Balance Across Lifecycle Stages ("Developing Winners" vs. "Cash Generators"):**

Allows top management to see at a glance whether the corporation has enough young SBUs in the Development and Growth stages (future winners) financed by strong SBUs in the Maturity stage (cash cows), avoiding an aging portfolio trapped in Decline.

• **3. Tailored Strategic Prescriptions for Weak/Average Units:**

Shows whether a weak or average SBU in the Growth/Shakeout stage should receive turnaround investment to build share or be divested before industry maturity locks in market shares.

• **4. Dual Visual Representation of Industry Size and SBU Market Share:**

By plotting industry size (circle area) and SBU share (shaded wedge), executives can simultaneously assess external market potential and internal competitive standing.

Group D: Comprehensive Answer Questions [20 Marks]

Q22. Case Study: Apple Inc. (Steve Jobs, MobileMe Crisis & Tim Cook's Operations) Apple Computer, Inc. (incorporated 1977 by Steve Jobs, Steve Wozniak, and Ronald Wayne) was renamed Apple Inc. in 2007 as focus shifted to consumer electronics (iPhone, iPad, Mac, iPod, Apple Watch, Apple TV; iOS, OS X, watchOS, tvOS; App Stores). Jobs positioned Apple as a 'value leader' not a price leader, and a premium manufacturer of innovative products. In 2008, Apple launched the MobileMe email/cloud service alongside iPhone, which became dubbed 'MobileMess' due to lost emails and faulty synchronization, prompting Jobs's famous Town Hall auditorium accountability meeting. Earlier in March 1998, Jobs recruited Tim Cook as SVP of Worldwide

Operations for his expertise in Just-In-Time (JIT) manufacturing. Jobs created a unique culture of perfectionism, innovation, and meeting productivity. a) Point out the main pricing strategy that Steve Jobs wanted to implement. b) What strategy did Apple Company apply to bring electronic goods in the market? c) What was the reason behind the 'Mobile-Mess'? Whom would you blame and why? d) What were the overall secrets of the Company's success? What suggestion would you like to give to Apple Company to improve its efficiency and performance?

Answer (Comprehensive Case Study Solution):

a) Main Pricing Strategy That Steve Jobs Wanted to Implement:

• Steve Jobs implemented a **Value-Based Premium Pricing Strategy (coupled with Price Skimming)**, explicitly positioning Apple as a **"Value Leader" rather than a "Price Leader."**

• Instead of competing on low prices or compromising product integrity just to cut costs, Jobs believed customers would gladly pay a premium price for breakthrough innovation, superior industrial design, seamless hardware-software integration, and user gratification.

b) Strategy Applied by Apple to Bring Electronic Goods into the Market:

• **1. Business-Level Generic Strategy — Broad Differentiation & Product Innovation Strategy:**

Apple differentiated its products through proprietary operating systems (iOS, macOS, watchOS, tvOS), iconic minimalist design, and a closed, seamless ecosystem (App Store) that created high customer loyalty and high switching costs.

• **2. Corporate-Level Strategy — Related (Concentric) Diversification Strategy:**

In 2007, Apple strategically redefined its business scope (renaming itself from *Apple Computer, Inc.* to *Apple Inc.*), leveraging its core competencies in microprocessor/software design and user interface to diversify from Mac computers into iPod, iPhone, iPad, Apple Watch, and Apple TV.

• **3. Functional Operations Strategy — Just-in-Time (JIT) & Global Supply Chain Optimization:**

By recruiting Tim Cook in 1998, Apple outsourced routine manufacturing, slashed inventory holding periods through Just-in-Time (JIT) inventory management, and locked in exclusive component suppliers.

c) Reason Behind the "Mobile-Mess" (2008 MobileMe Failure), Whom to Blame, and Why:

• **Reasons Behind the "Mobile-Mess":**

1. *Core Competency Mismatch (Hardware vs. Cloud Services):* While Apple excelled at designing standalone hardware devices and operating systems, it lacked deep operational experience in running always-on, real-time cloud synchronization servers (unlike BlackBerry's enterprise servers).
2. *Rushed Simultaneous Launch & Inadequate Stress-Testing:* Launching MobileMe simultaneously with the iPhone 3G and App Store overwhelmed Apple's servers, causing lost emails, sync failures, and downtime.
3. *Siloed Communication and Lack of Pre-Launch Escalation:* The MobileMe engineering team failed to flag readiness issues before launch.

• **Whom to Blame and Why:**

1. *Primary Operational Blame — The MobileMe Project Leadership & Quality Assurance Team:* For releasing an insufficiently stress-tested cloud service that failed to meet Apple's standard of perfection and tarnished the iPhone launch.
2. *Shared Strategic Blame — Top Management's Unrealistic Deadline Pressure:* Setting a rigid, simultaneous launch deadline without phased beta testing contributed to the premature rollout. (However, Jobs's decisive post-mortem accountability meeting successfully transformed the failure into today's highly reliable **iCloud**).

d) Overall Secrets of Apple's Success & Strategic Suggestions to Improve Future Efficiency and Performance:

• **Overall Secrets of Apple's Success:**

1. *Relentless Customer-Centric Innovation & Ecosystem Lock-In:* Seamless integration of hardware (iPhone, Mac, Watch), software (iOS, macOS), and services (App Store, iCloud) that anticipates customer desires.
2. *Complementary Strategic Leadership (Visionary + Operational Genius):* Combining Steve Jobs's visionary product design and marketing brilliance with Tim Cook's world-class Just-in-Time (JIT) supply chain and operational efficiency.
3. *Strong, Internalized Corporate Culture:* A culture of perfectionism, accountability, design excellence, and high productivity.

• **Suggestions to Improve Future Efficiency and Performance:**

1. *Geographic Supply-Chain Diversification:* Reduce over-concentration of manufacturing in a single country by expanding production across India, Vietnam, and other hubs to mitigate geopolitical and supply-chain disruption risks.
2. *Phased Beta Testing for Cloud and AI Services:* Use staged rollouts and rigorous cloud stress-testing for new AI/software services to prevent any repeat of the "MobileMess" incident.
3. *Expanding High-Margin Digital Services & Emerging-Market Reach:* Grow recurring revenue in cloud, fintech (Apple Pay), health-tech, and AI while offering trade-in/financing programs in fast-growing emerging markets.